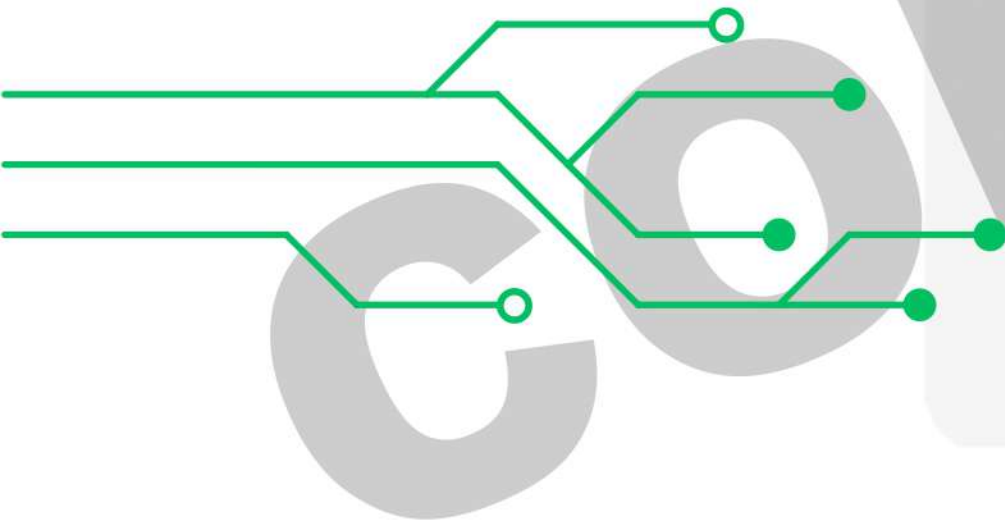




GeM
Government
e Marketplace

SENANITECH

ELECTRONICS DESIGN AND MANUFACTURING SERVICE

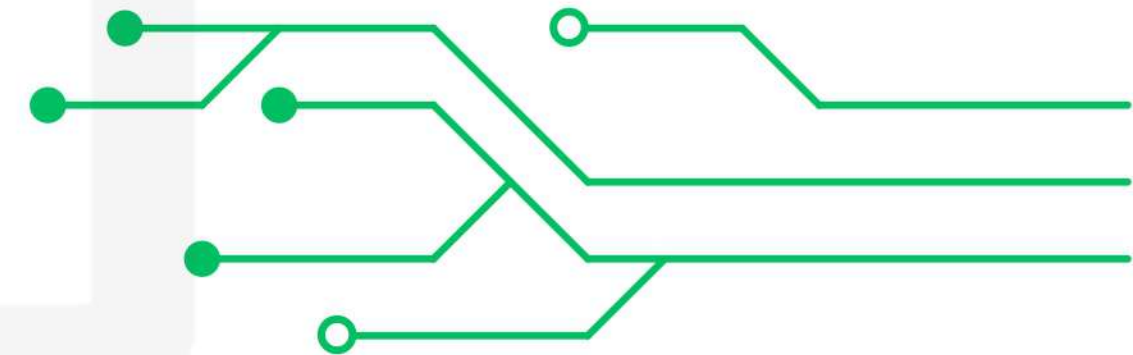


ABOUT US

Founded in 2021, we are a dynamic Electronic Engineering and R&D Services company driven by a passion for innovation and excellence in technology. With a strong focus on delivering cutting-edge solutions, we specialize in hardware design, PCB development, and comprehensive research and development services tailored to meet the specific needs of our clients.

FOCUS AREAS

- Hardware Design
- PCB Design
- R&D Services
- Embedded Development

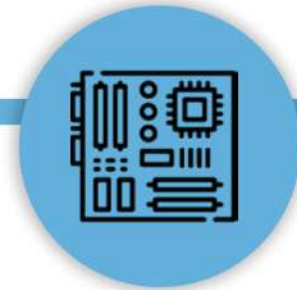


OUR SERVICES



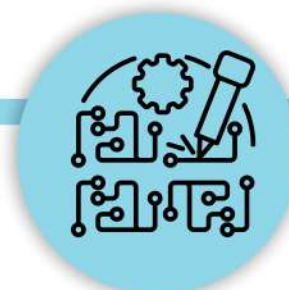
CONCEPT

- Concept Validation
- Product Specification Development
- System Architecture



HARDWARE DESIGN & FIRMWARE DEVELOPMENT

- HW High Level Design
- HW Low Level Design
- Component Engineering
- Board Bring-Up
- Testing
- Firmware Development
- Driver Development



PCB DESIGN & SI/PI SIMULATION

- PCB Layout Design
- Signal / Power / Thermal Integrity Analysis
- High-Speed Design (HDI and ATI Boards)



ELECTRONICS MANUFACTURING SERVICE

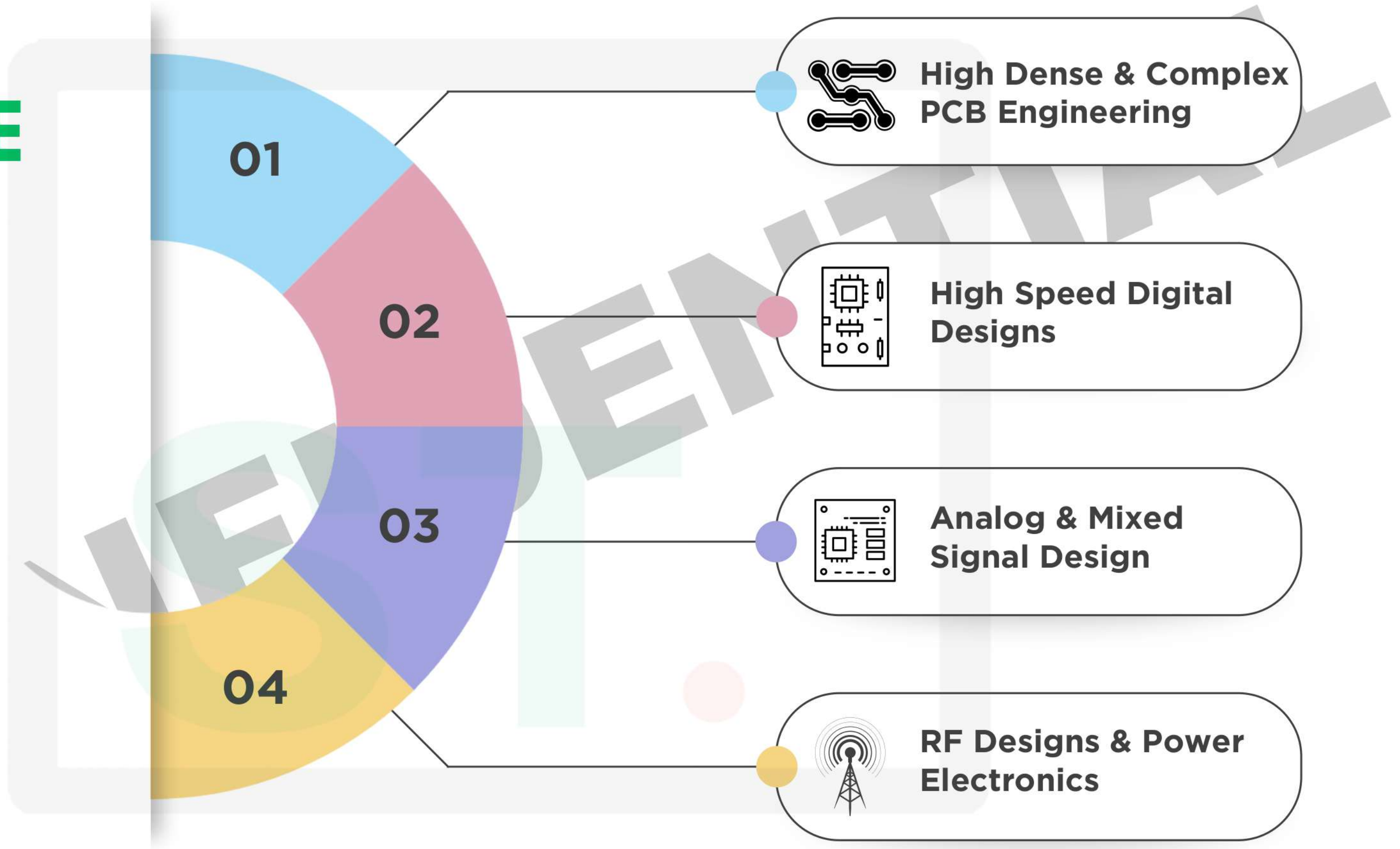
- PCB Manufacturing (Thru Partners)
- PCB Assembly
- Power ON Testing
- Box Building
- Cable Harness



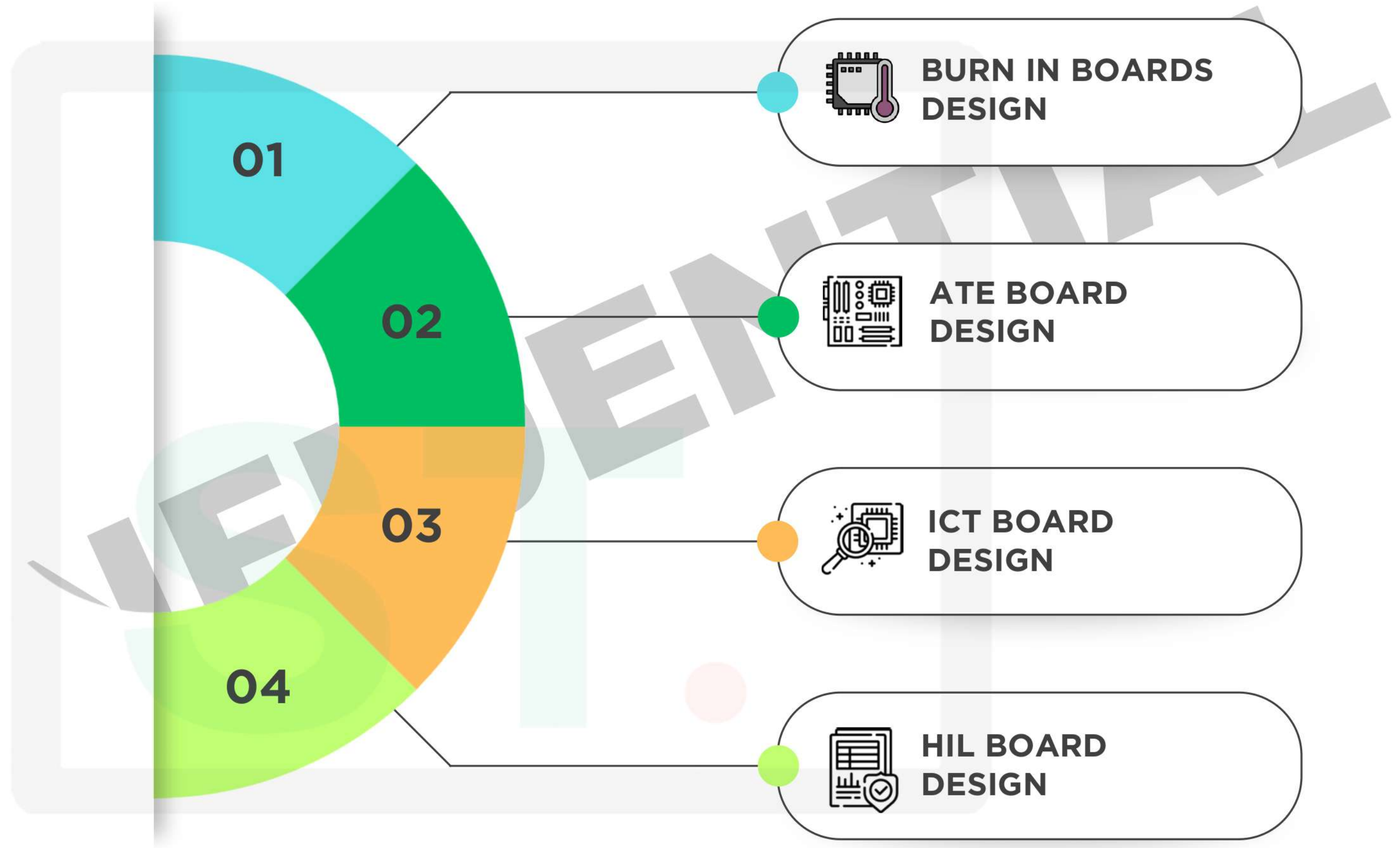
REGULATORY COMPLIANCE

- IEC 60601-1:2024
- ISO 13485

DESIGN EXPERTISE



PCB DESIGN



TOOLS EXPERTISE

PCB Layout

Altium Designer, Allegro X Design ,
Eagle Autodesk, Kicad.

Schematic Entry

Altium Designer, Allegro X Design ,
Eagle Autodesk, Kicad.

Signal Integrity / Power Integrity

Keysight ADS PathWave, HFSS

Thermal Analysis

Ansys ICEPACK

CAM

CAM350 / UCAM

PACKAGE DESIGN

Allegro ADP

BURN-IN-BOARDS

Senanitech delivers end-to-end Reliability and Burn-In solutions, offering expert BIB design that maximizes test coverage at minimum cost. Our experienced team handles CAF, tight pitch, high power, de-rating, and custom circuitry needs – making us a trusted partner for semiconductor reliability challenges.

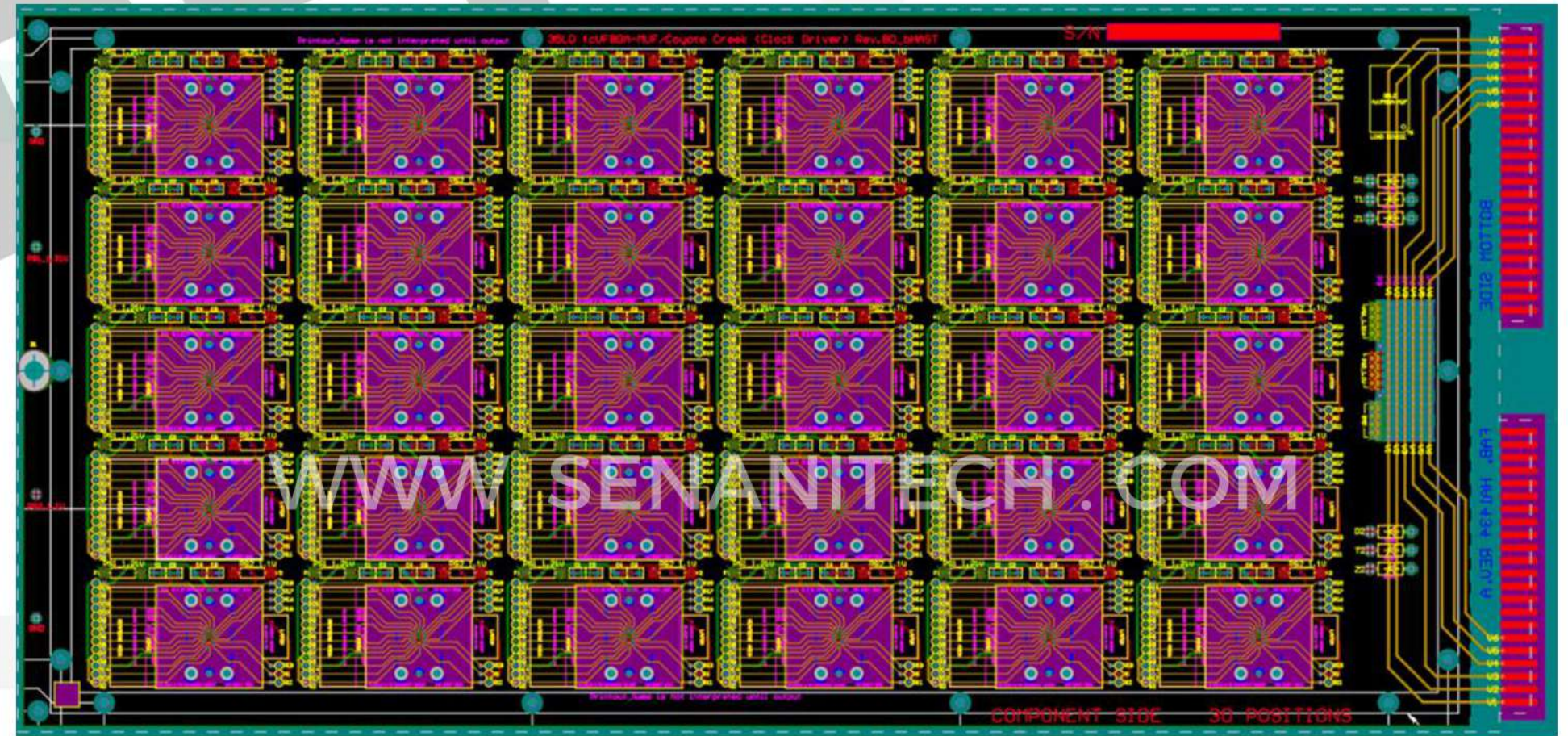
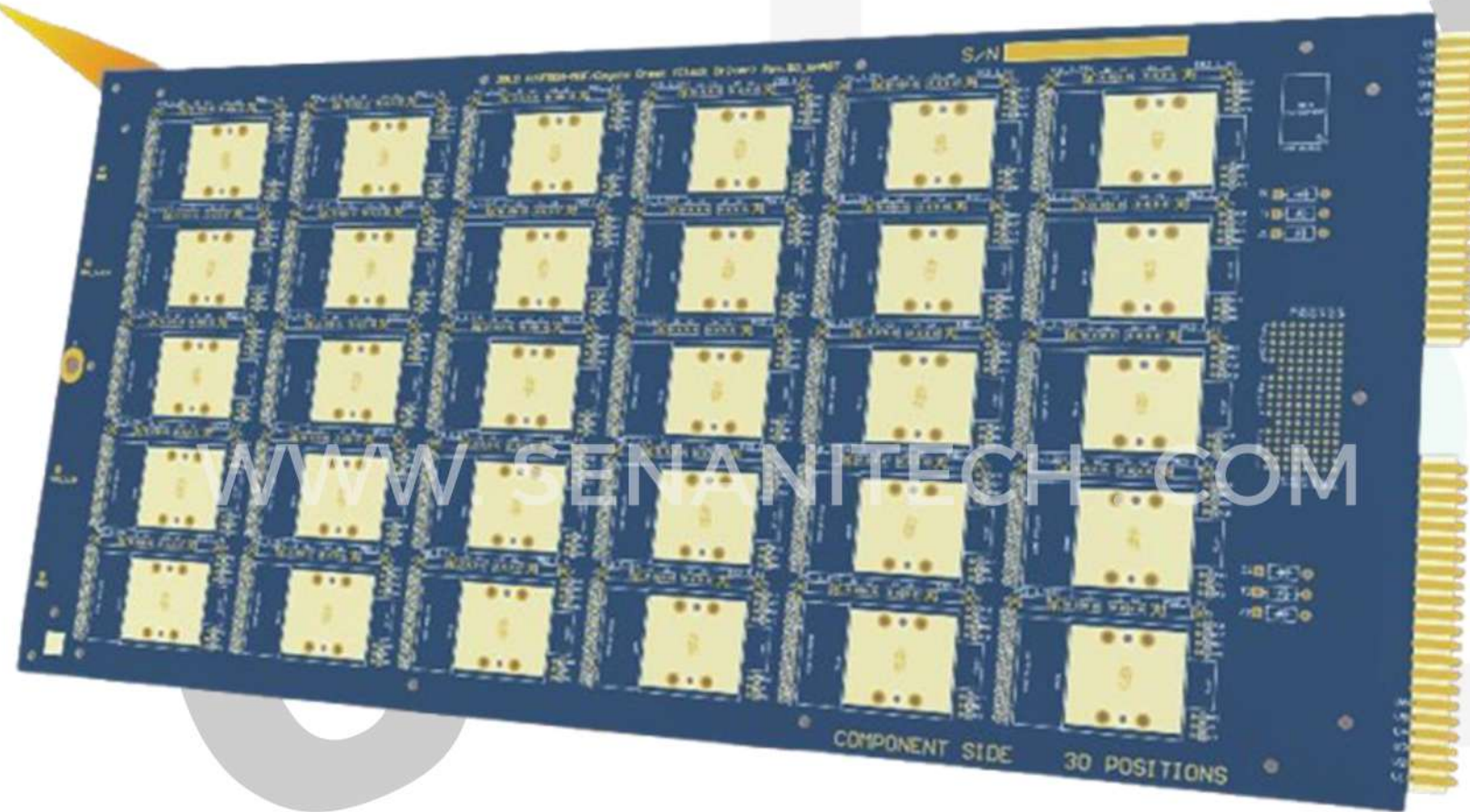
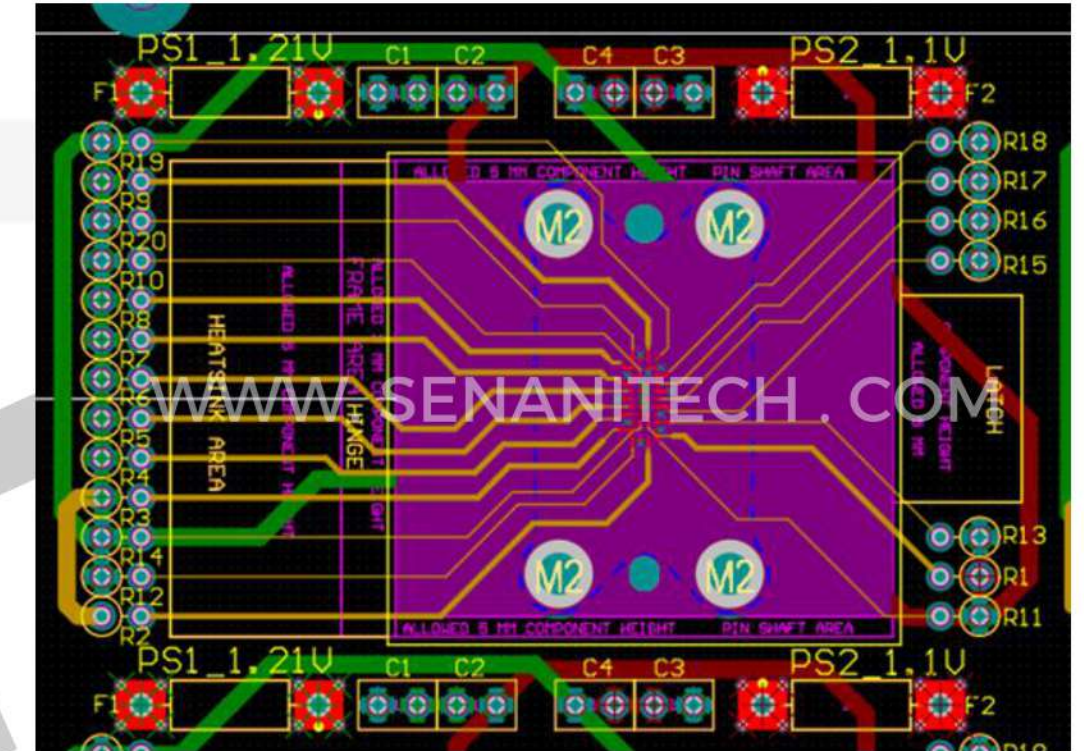
Types of Boards:

- Burn-in Boards for All Systems (Including High-Power BIB's)
- HAST Boards
- HTOL- Infinity i160/ Infinity i8160/ Tahoe
- Test Fixtures for Failure Analysis
- THB (Temperature-Humidity-Bias)
- Special Application Boards



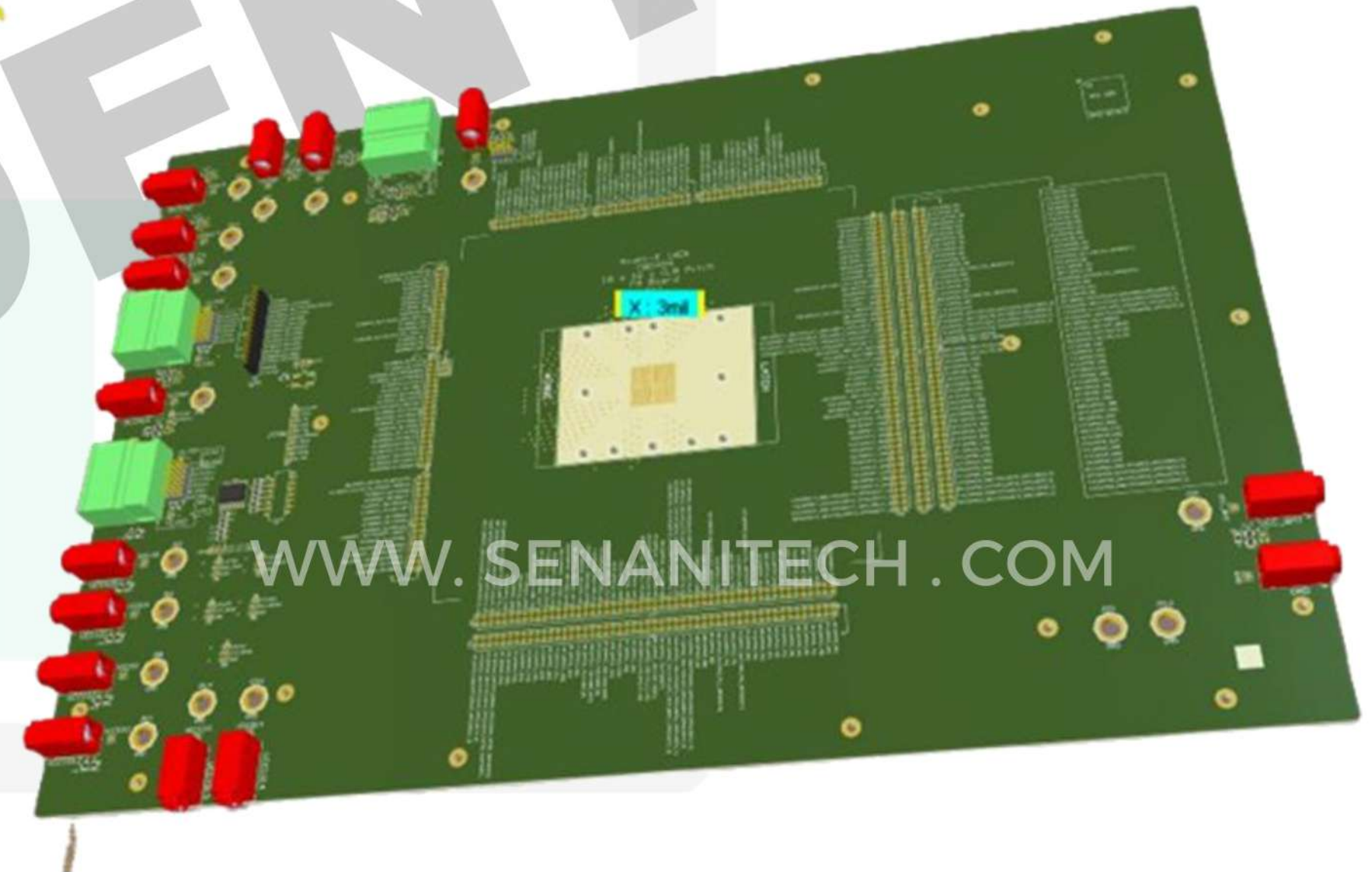
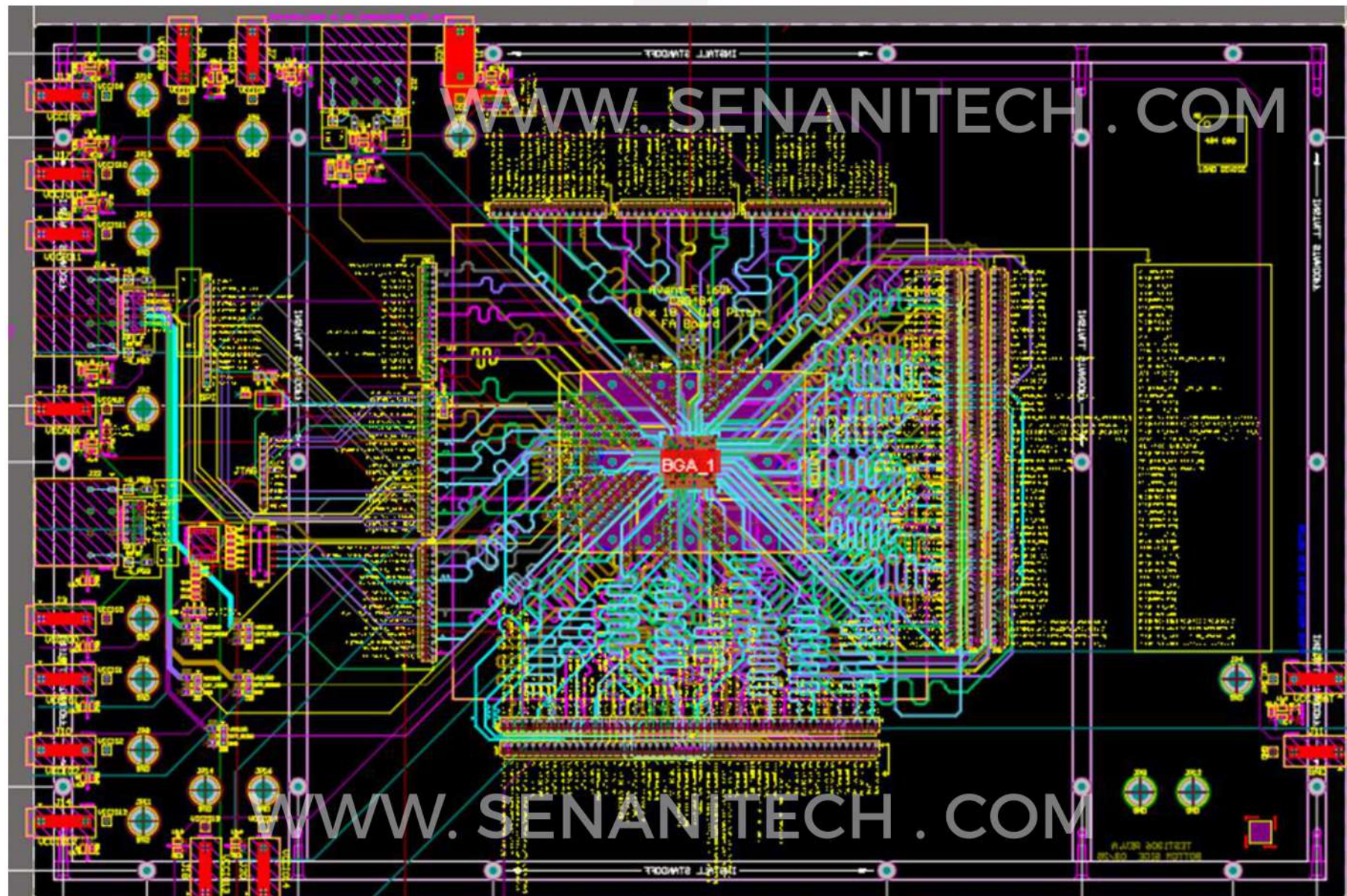
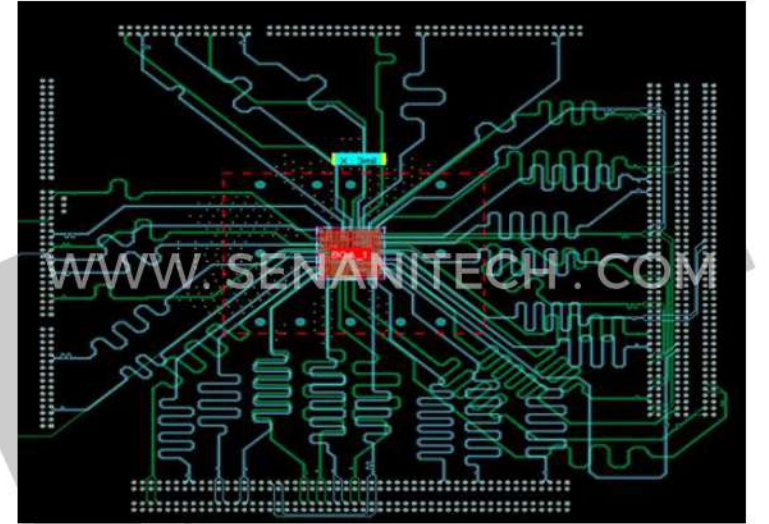
Hast Board

- Up to 10-Layer Stack up: Uses Polyimide
- 30-Cell Optimized 30-DUT matrix placement to ensure 100% uniform.
- Dog-Bone Via Fan-out: Precision dog-bone escape routing to prevent solder wicking
- HAST Leakage Control: Strict 15-20 mil isolation spacing
- Develop a customized socket footprint.



Hast Board

- **Package Detail:** Up to 1156-pin BGA | Ultra-High Density | Fine-Pitch.
- **Power Design:** Up to 20+ Independent Rails | High-Current Plane Segmentation.
- **Routing:** Phase-Matched Differential Pairs | Within-Group Length Tuning.
- **Fan-out:** 100% Net Accessibility | Full Diagnostic Fault Isolation (FI).
- **Stack up:** Up to 22-Layer Ultra-Dense Architecture | Advanced Impedance Control.

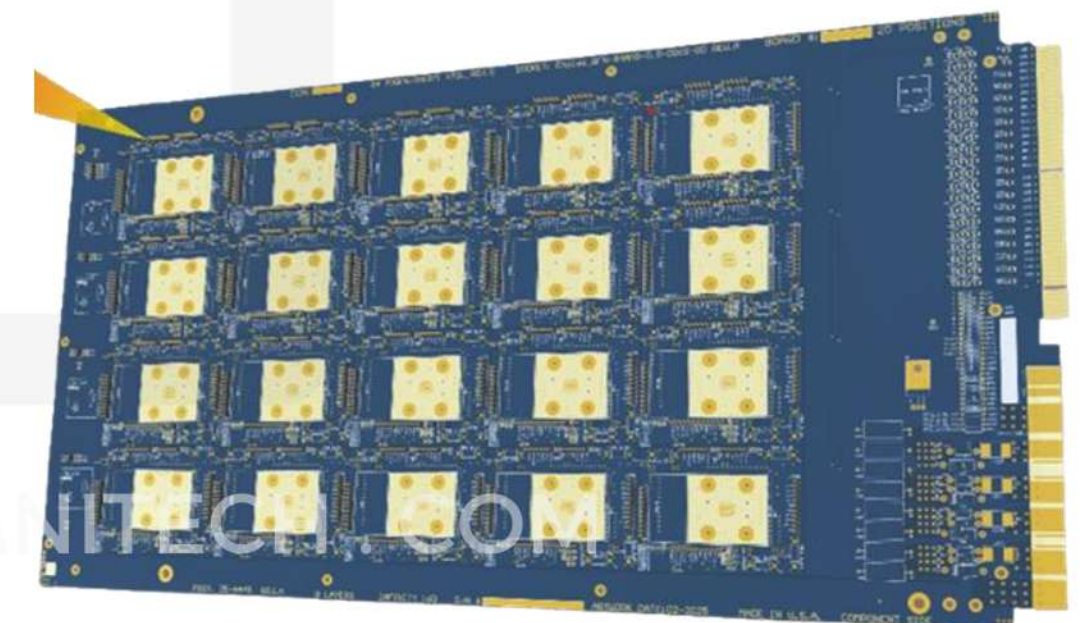
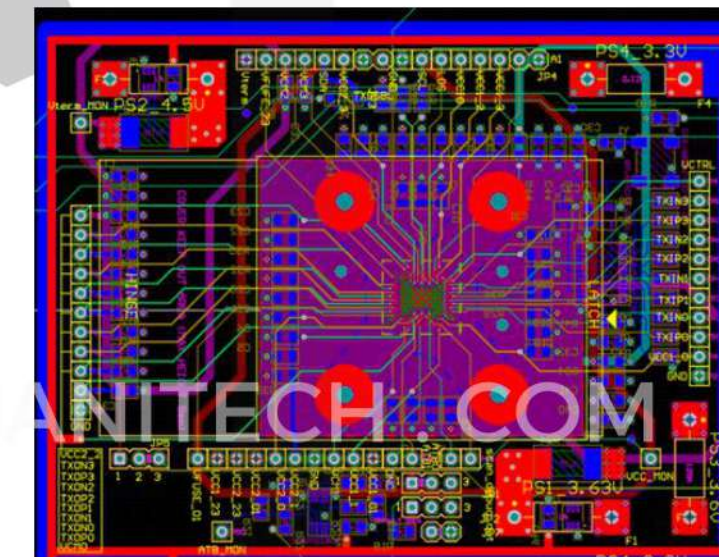
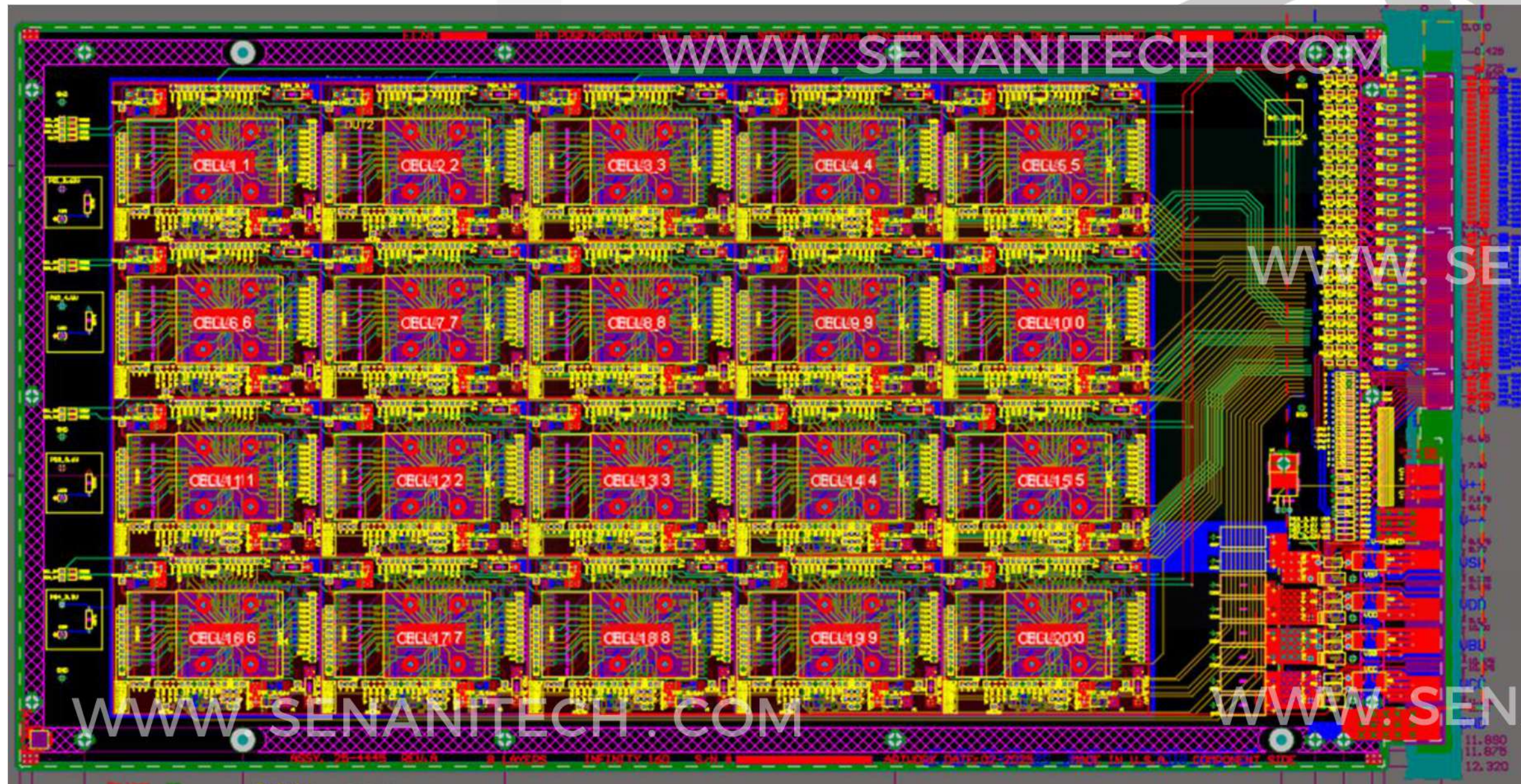


Infinity i160 & i8160 HTOL Platforms



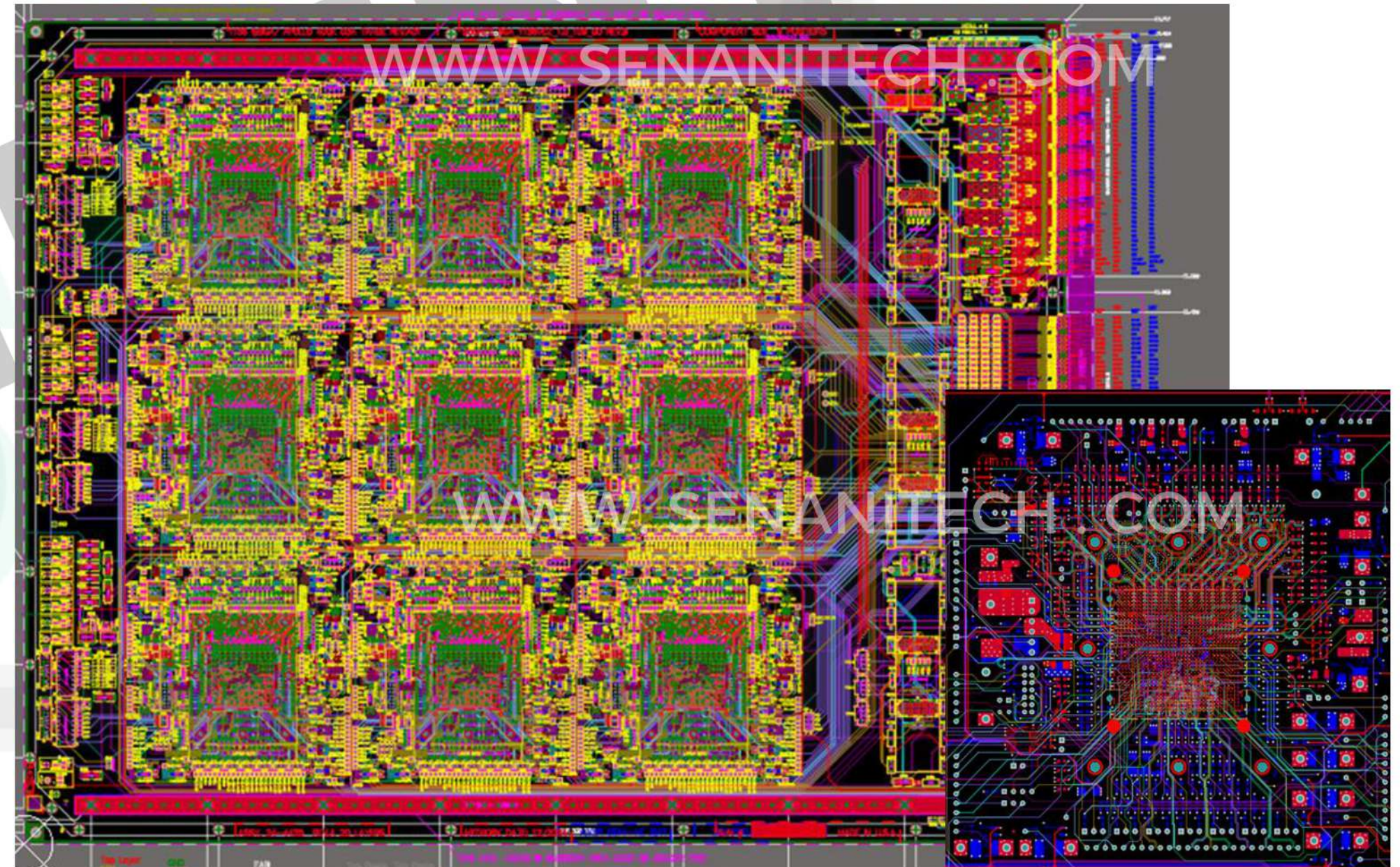
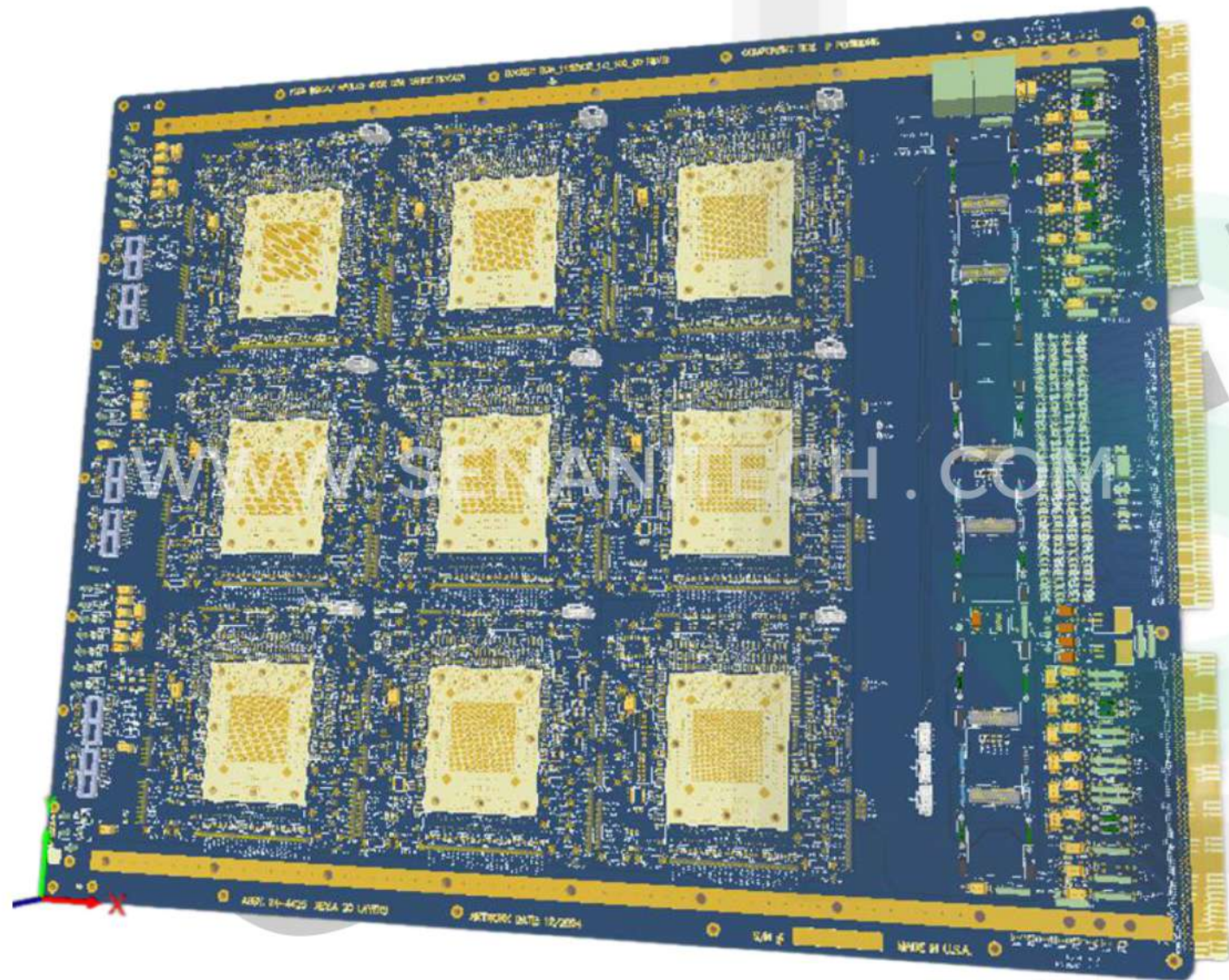
GeM
Government
e Marketplace

- **Package Detail:** 84 PDQFN | 30-Cell High-Density BIB Matrix.
- **Power Design:** i160 (4-Rail) | i8160 (8-Rail) | Integrated Per- Dut Power Monitoring.
- **Routing Strategy:** Precision Per-Row & Per-BIB isolation with Kelvin Sensing for voltage accuracy.
- **Fan-out Tech:** Dedicated Dog-Bone escape to maintain heat stress.
- **Critical Layering:** 8-Layer Ultra-Dense Stack up | Balanced copper distribution to prevent thermal warping.

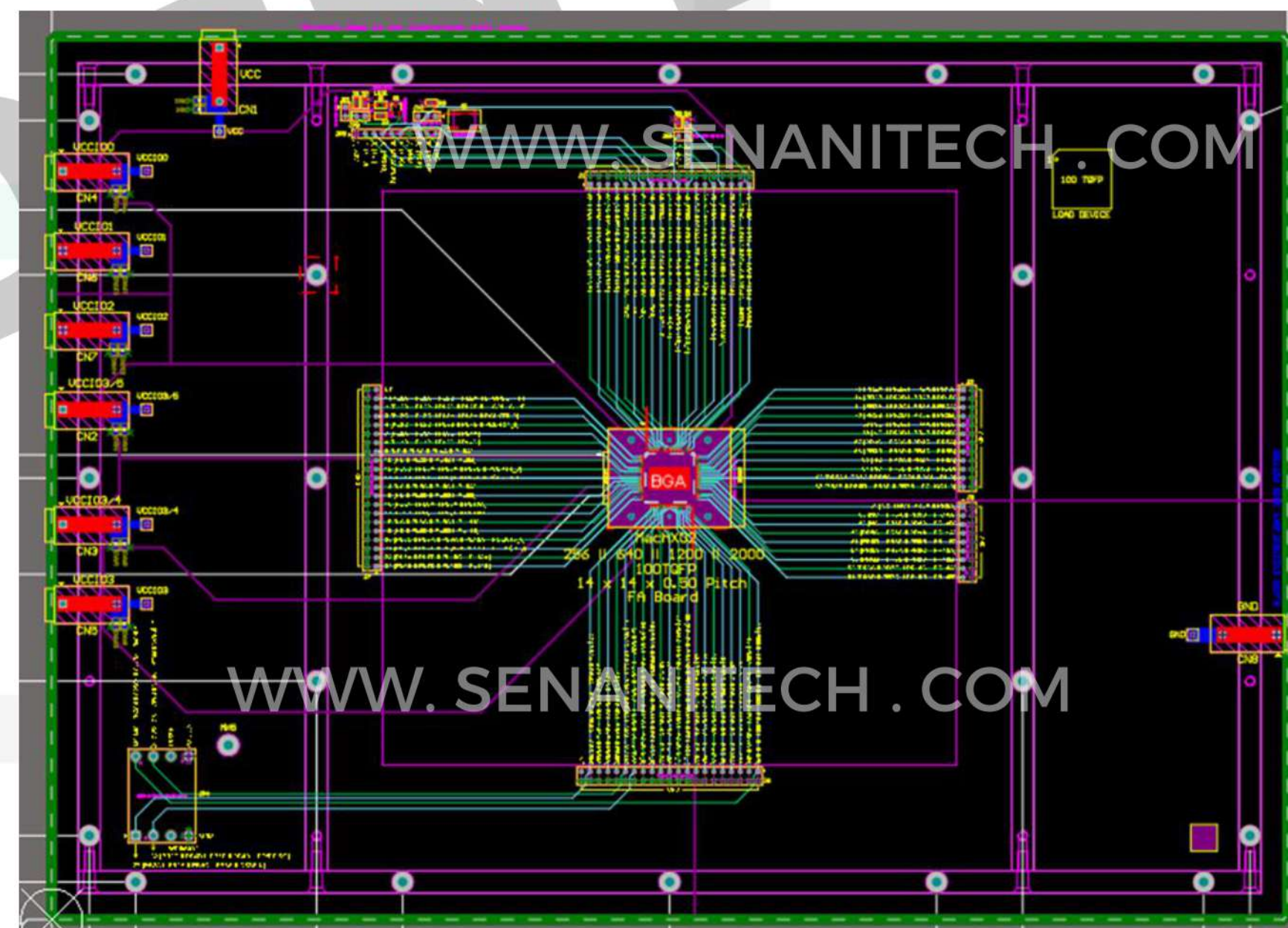
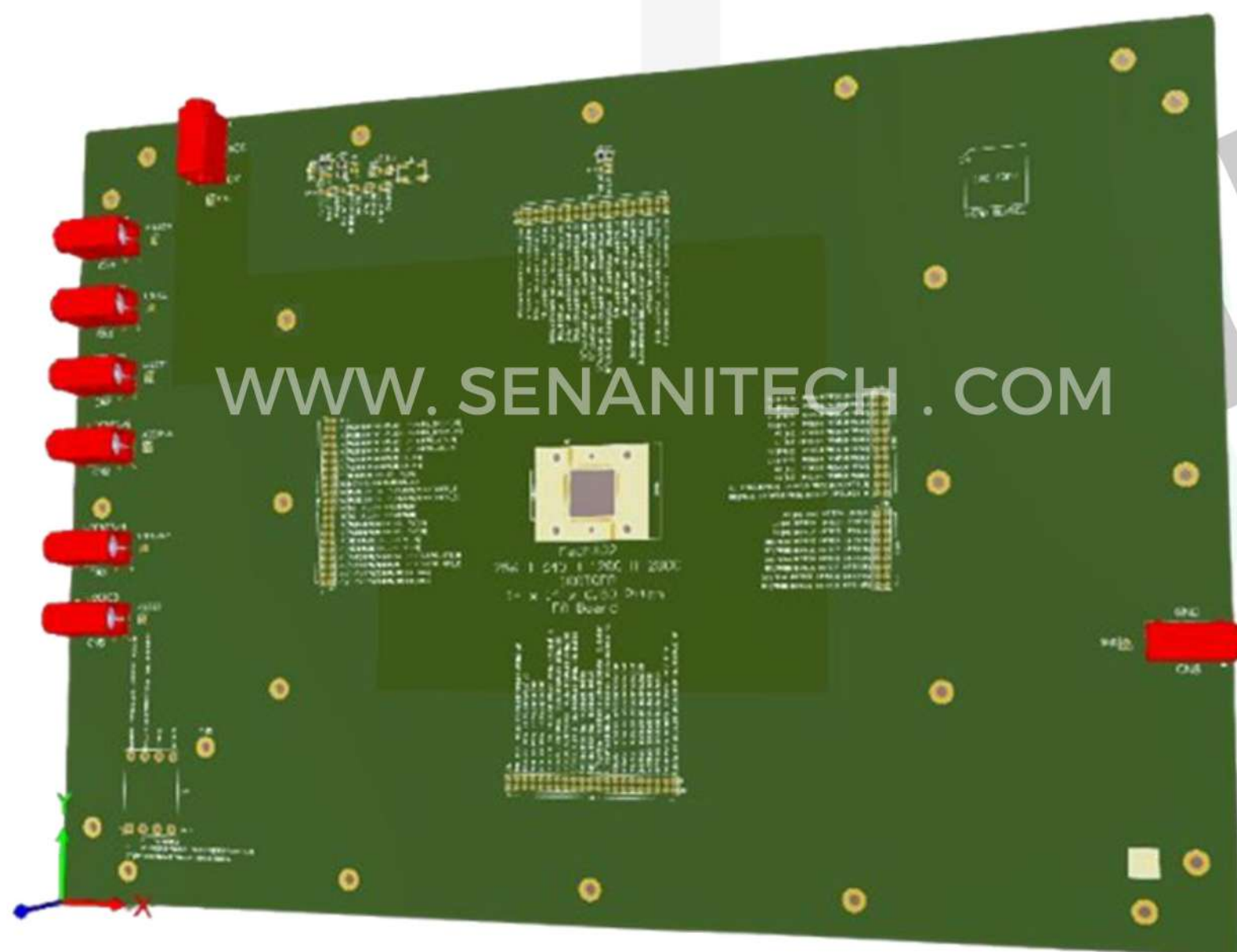


Tahoe High-Performance HTOL

- **Package Detail:** LFG1156 (1156-Pin) | 400K Logic Gate Density | 9-DUT (9-Socket) Matrix.
- **Power Design:** Ultra-High Current Distribution | Per- DUT Power Monitoring | Multi-Phase Voltage Rails.
- **Fan-out Tech:** High-Density Dog-Bone escape for 1.0mm pitch; designed for zero mechanical interference with socket hardware.
- **Critical Layering:** 22-Layer Ultra-Dense Stack up | Massive Copper planes for high-current thermal heat-sinking.

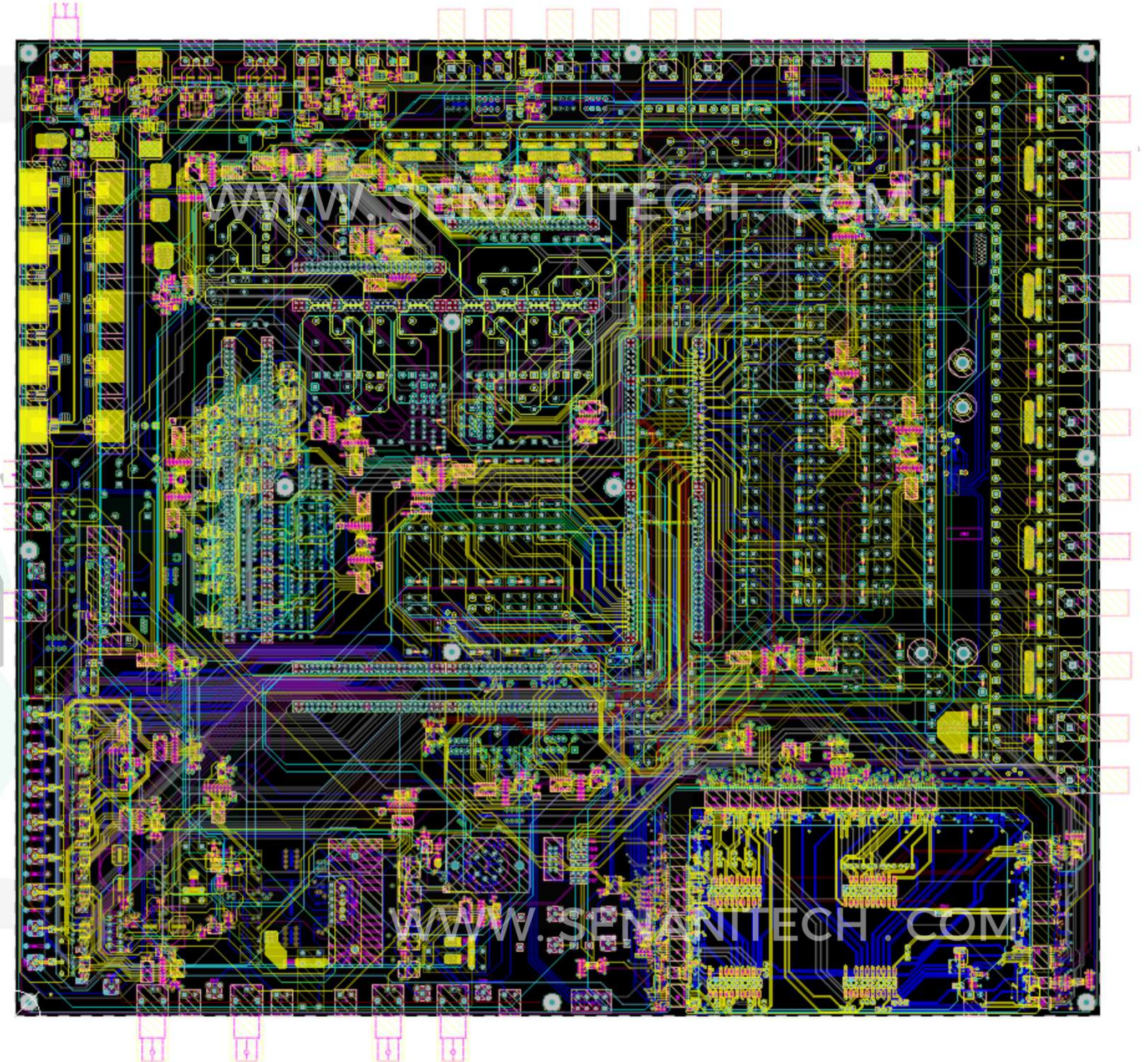
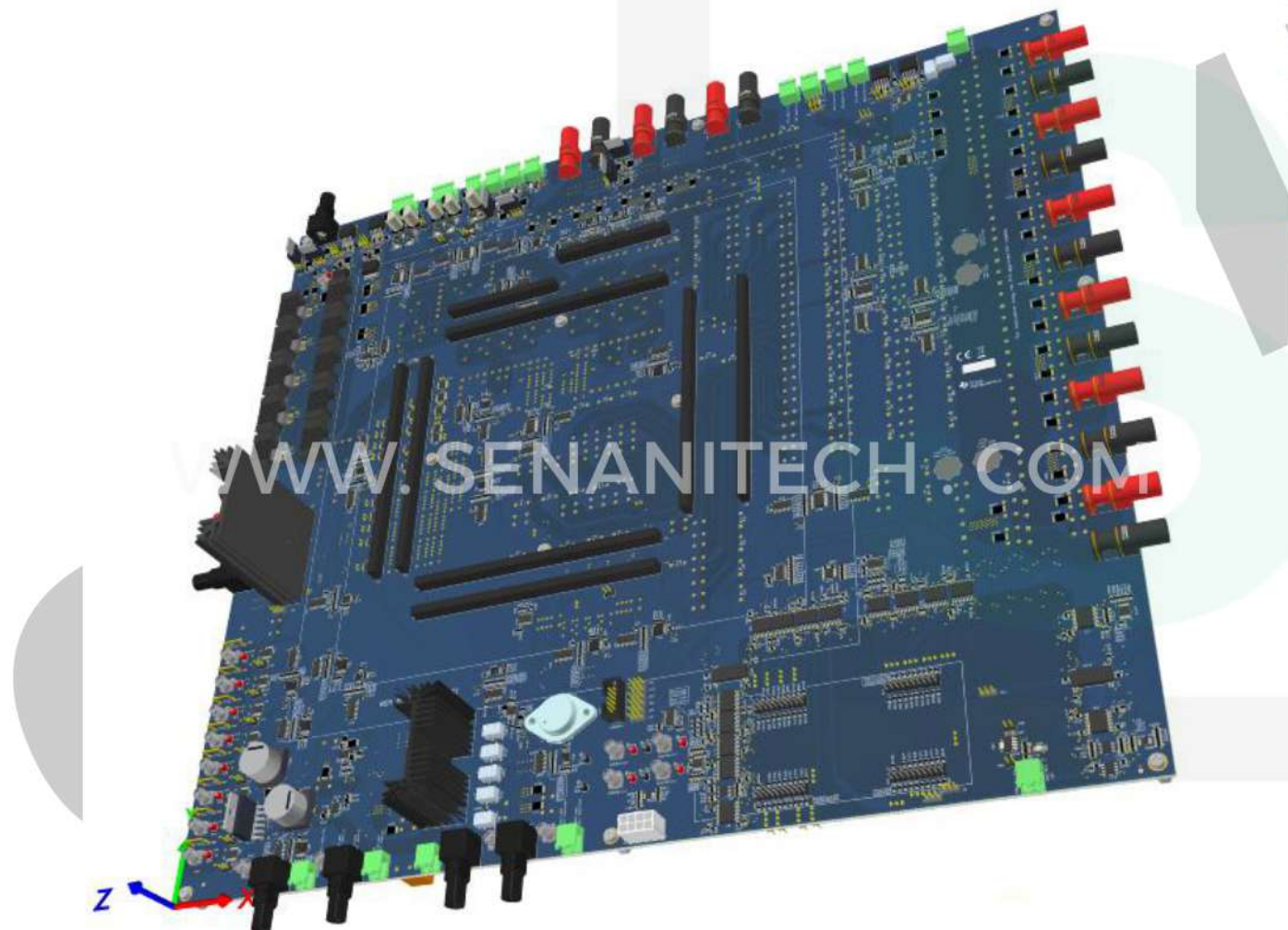


- **Package Detail:** Up to 1156-pin BGA | Ultra-High Density | Fine-Pitch.
- **Power Design:** Up to 10+ Independent Rails | High-Current Plane Segmentation.
- **Routing:** All the signals trace are having the same propagation delays
- **Fan-out:** 100% Net Accessibility | Full Diagnostic Fault Isolation (FI).
- **Stack up:** Up to 12-Layer Ultra-Dense Architecture | Advanced Impedance Control.



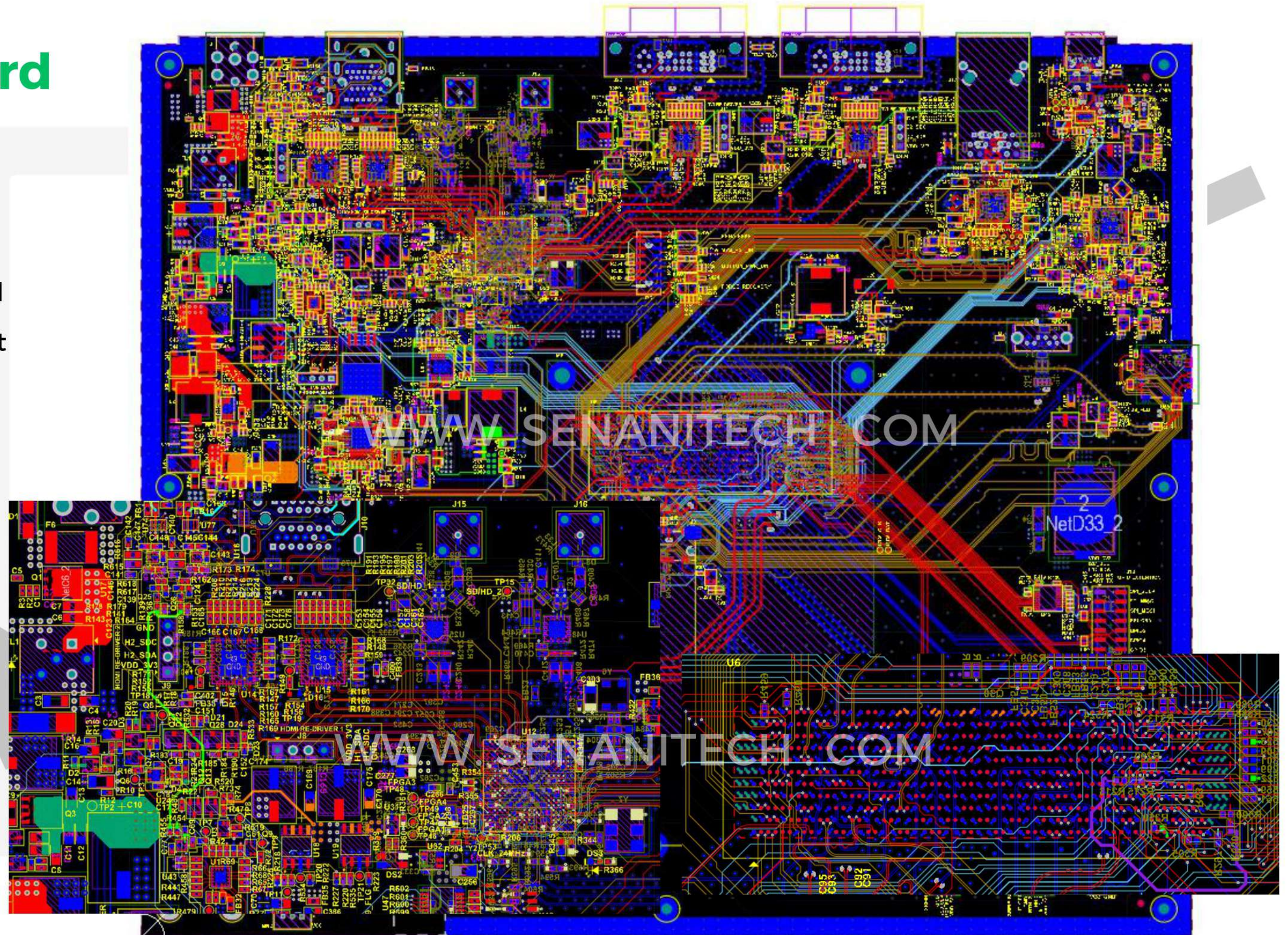
Industrial Automation Relay Board

- Master Relay Board
- 12 layers / 4597 Pins
- High Current
- Ultrahigh density
- Altium Designer
- Chipset - C2000



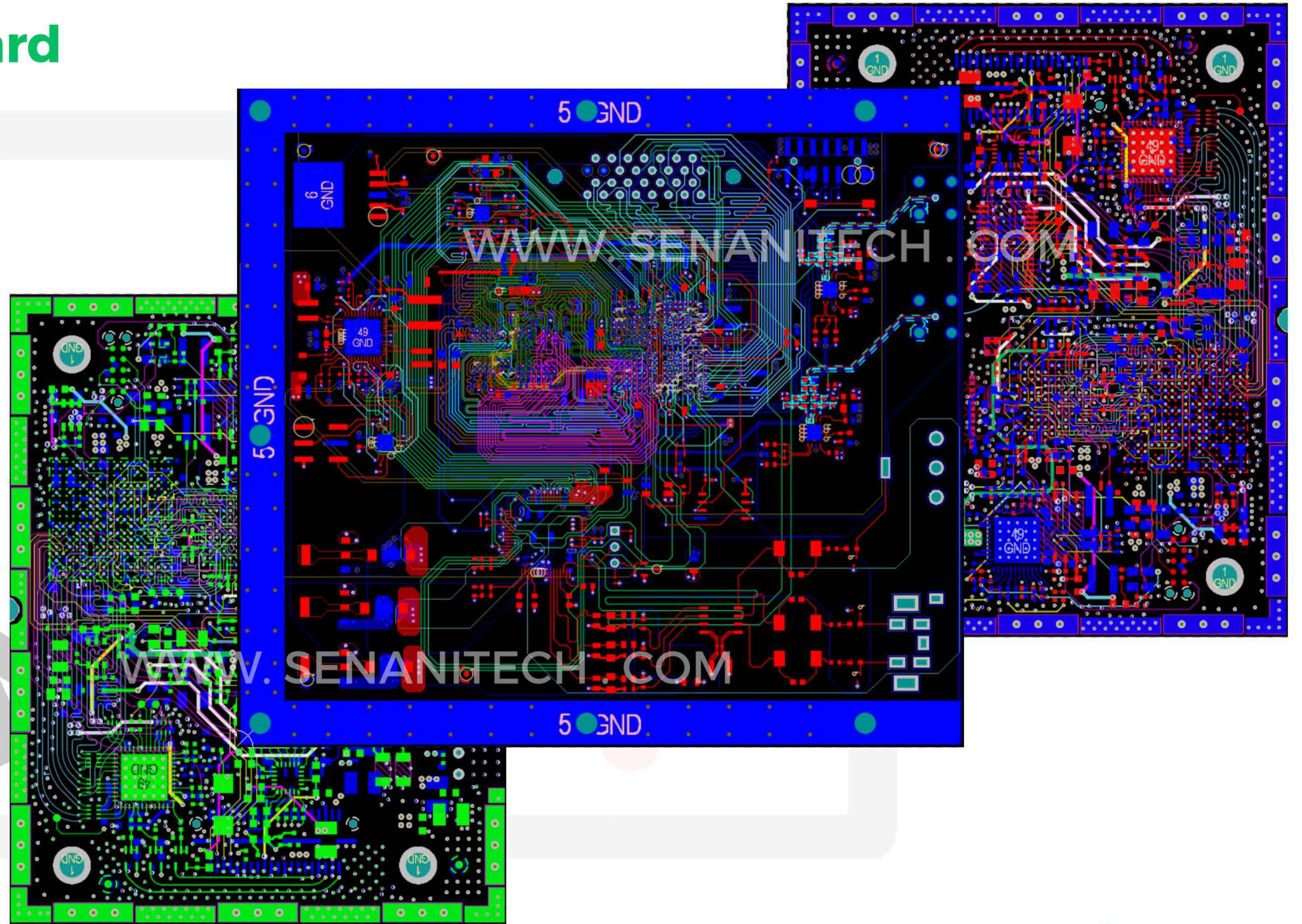
AGX Motherboard

- Commercial Application
- 16 layer
- HighSpeedDigital
- FPGA / nVidia Motherboard
- High-Speed Display Output
- 8326 pins
- Frequency 2.2GHz



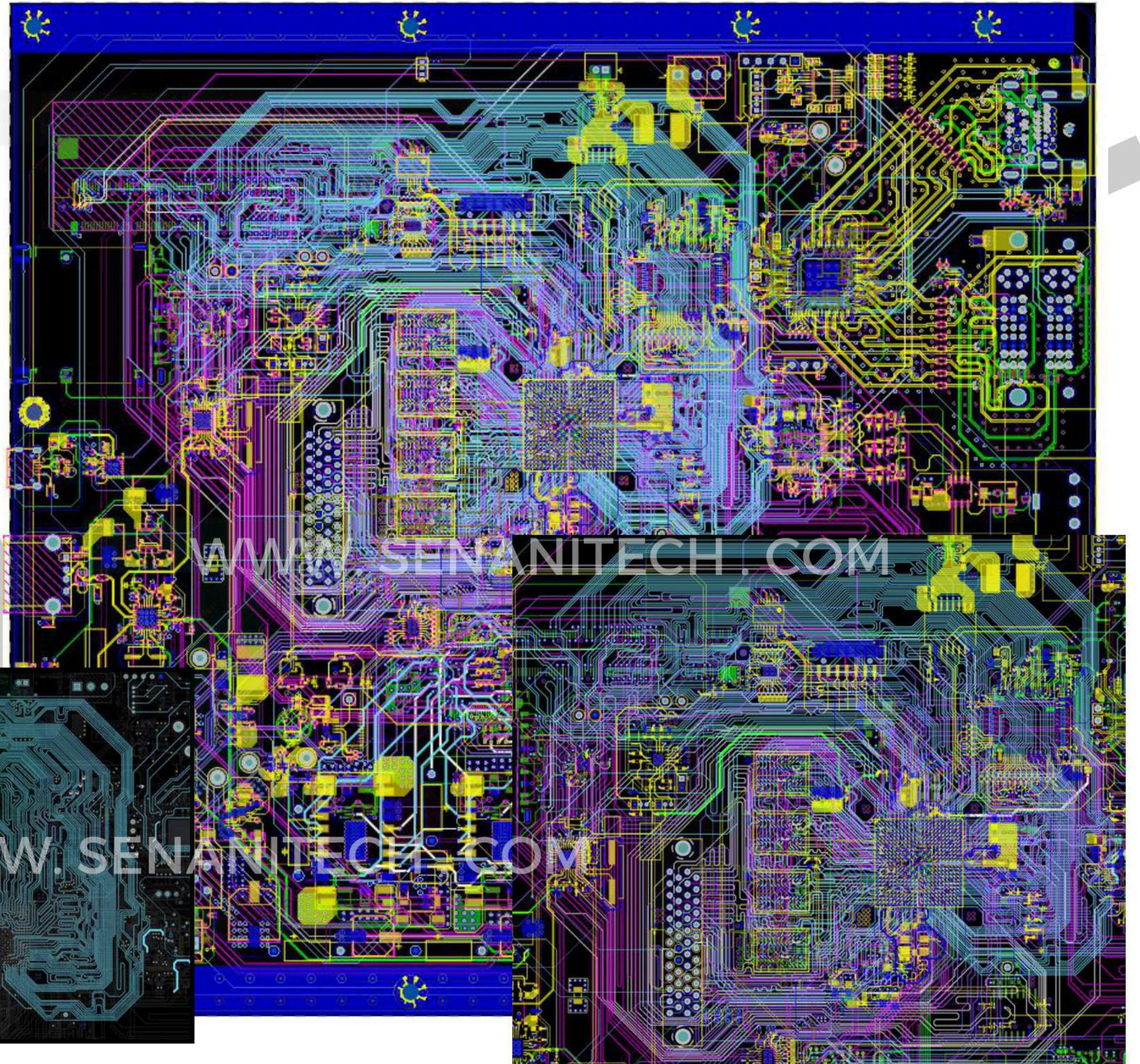
Rugged SBC Board

- Ultra-high density
- Military Application
- 14 layer
- HighSpeedDigital
- HDI-Blind/Buried Sets
- FPGA/LPDDR
- 12 Layer



FPGA Processor Board

- FPGA Processing Hardware
- Highspeed digital/HDMI/FPGA
- Altium Designer 21
- High Current
- Digital PowerTrain
- Total pincount:9360
- Impedance Requirement
- 4xDDR3L
- 10 Layer

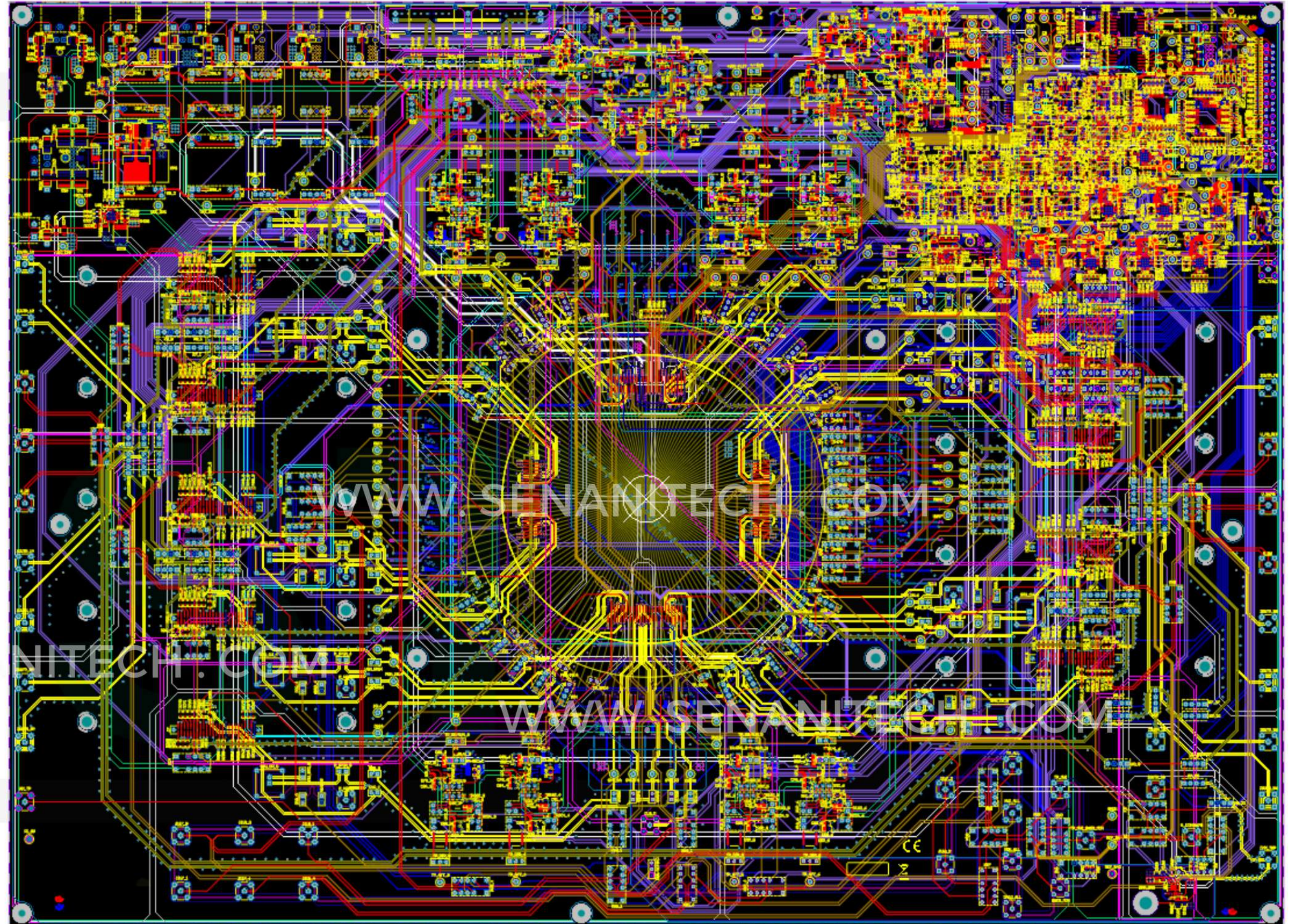
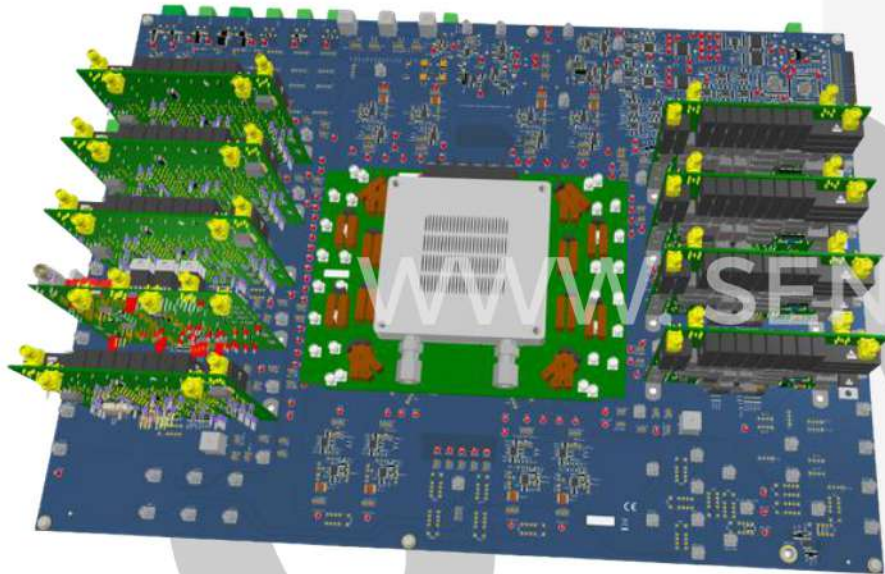


RIGID - FLEX PCBS

- Low density, Medical grade wireless.
- Wireless BLE Application
- 6 layer
- RF Based
- High-Volume product
- BLE-5

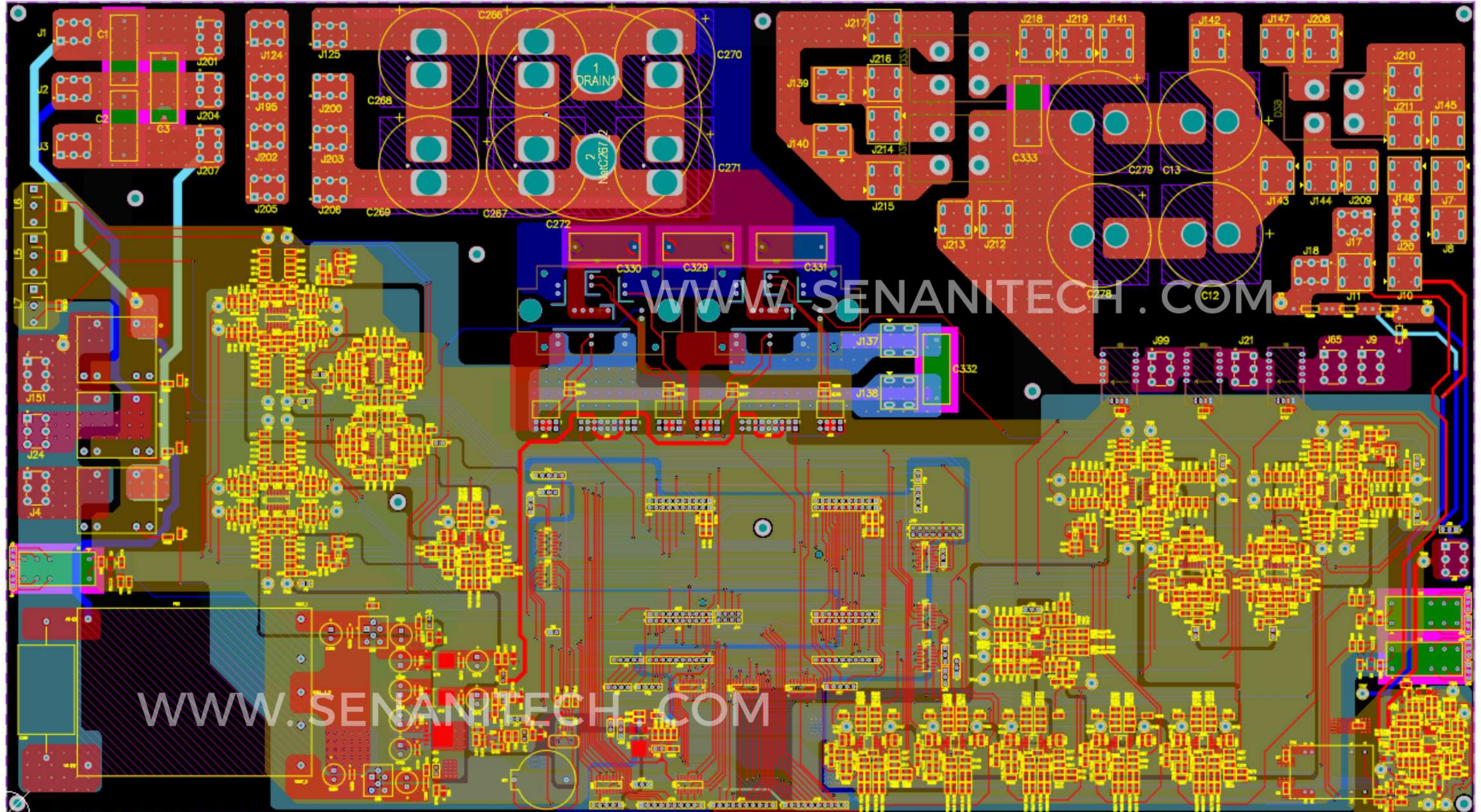
PMU Device

- High density
- High Current Board
- 18 layers
- Parametric measurement unit (PMU)
- 6532 pins



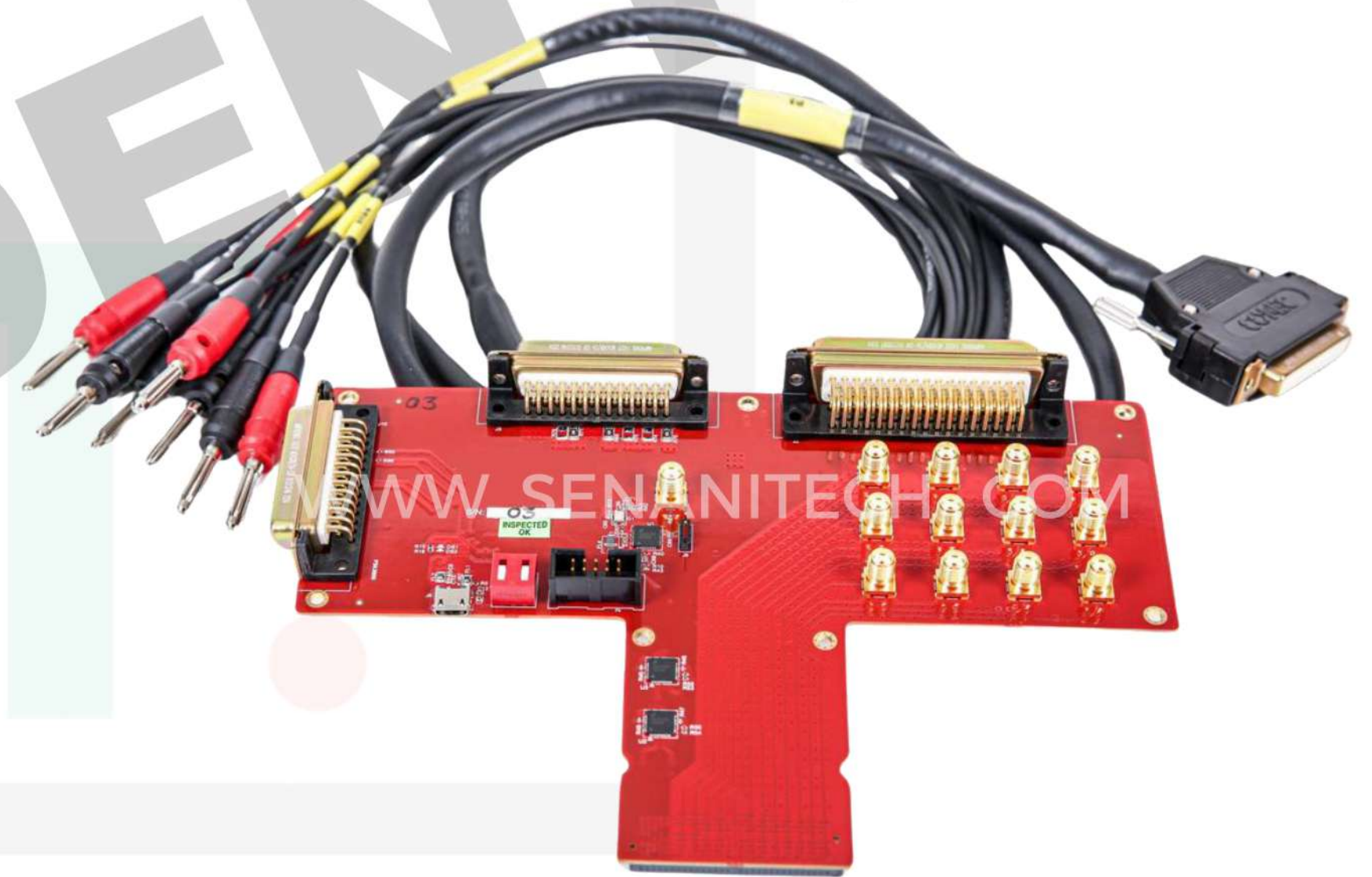
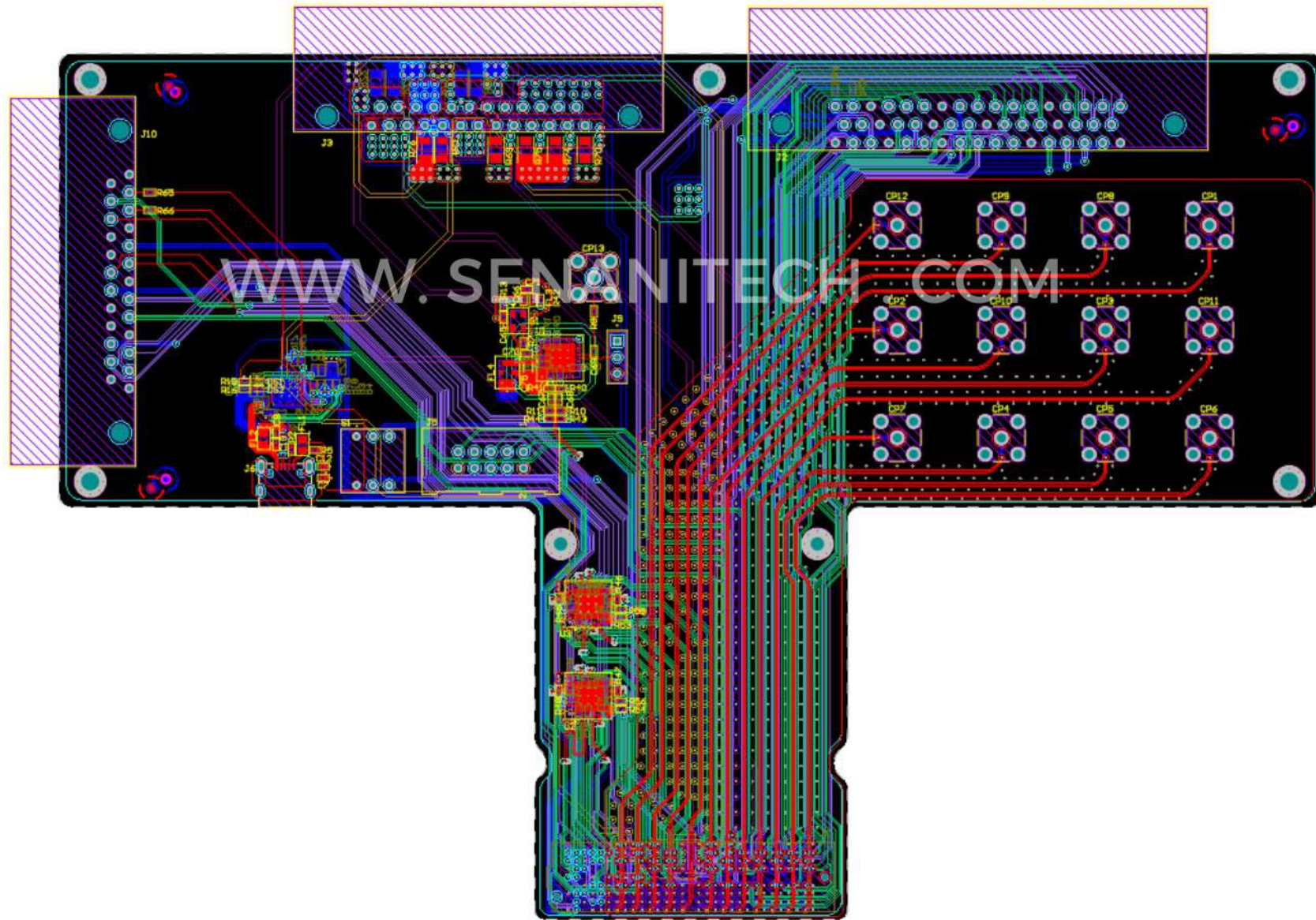
High Power Board - Traction Control

- HV Board
- 4 layers
- 120A
- 570mm X 370mm
- Automotive



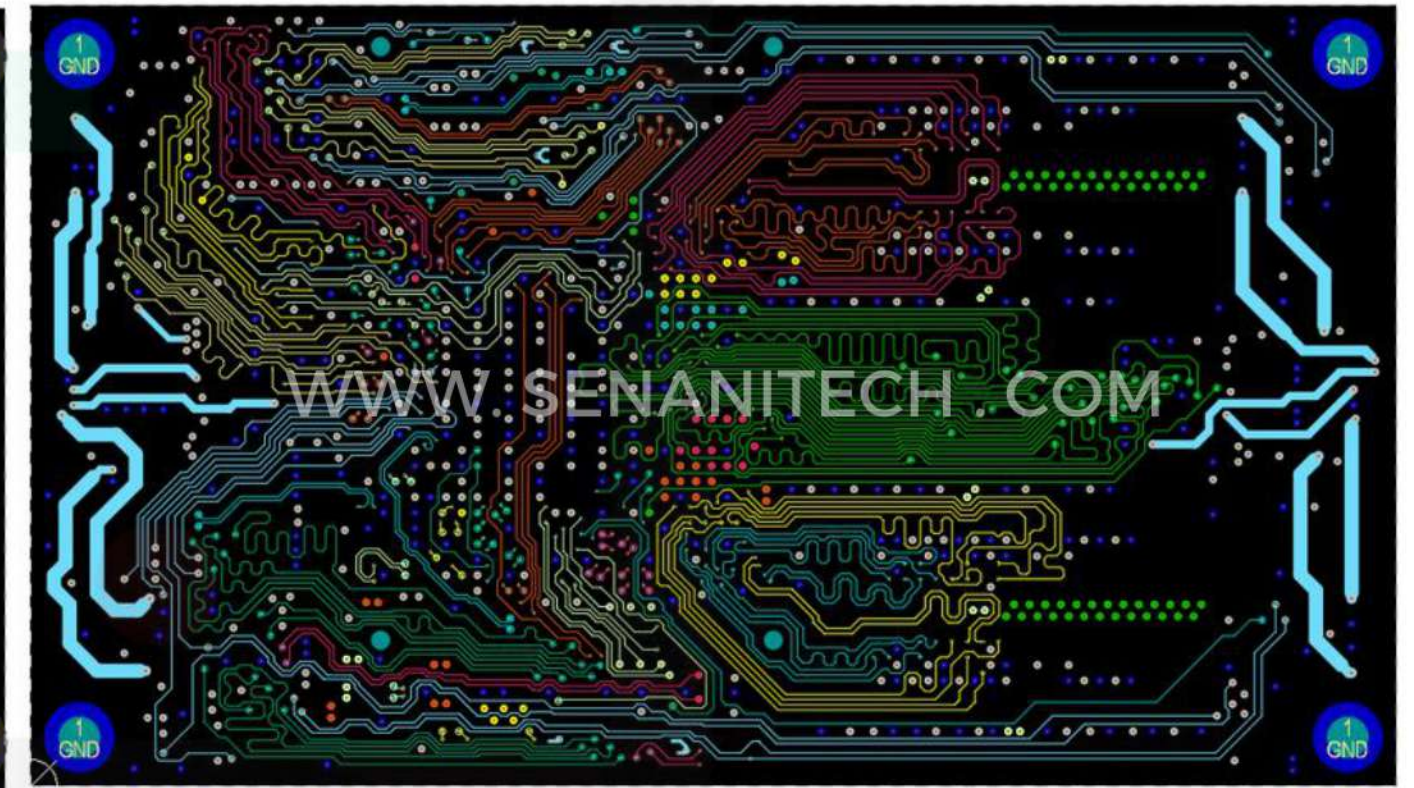
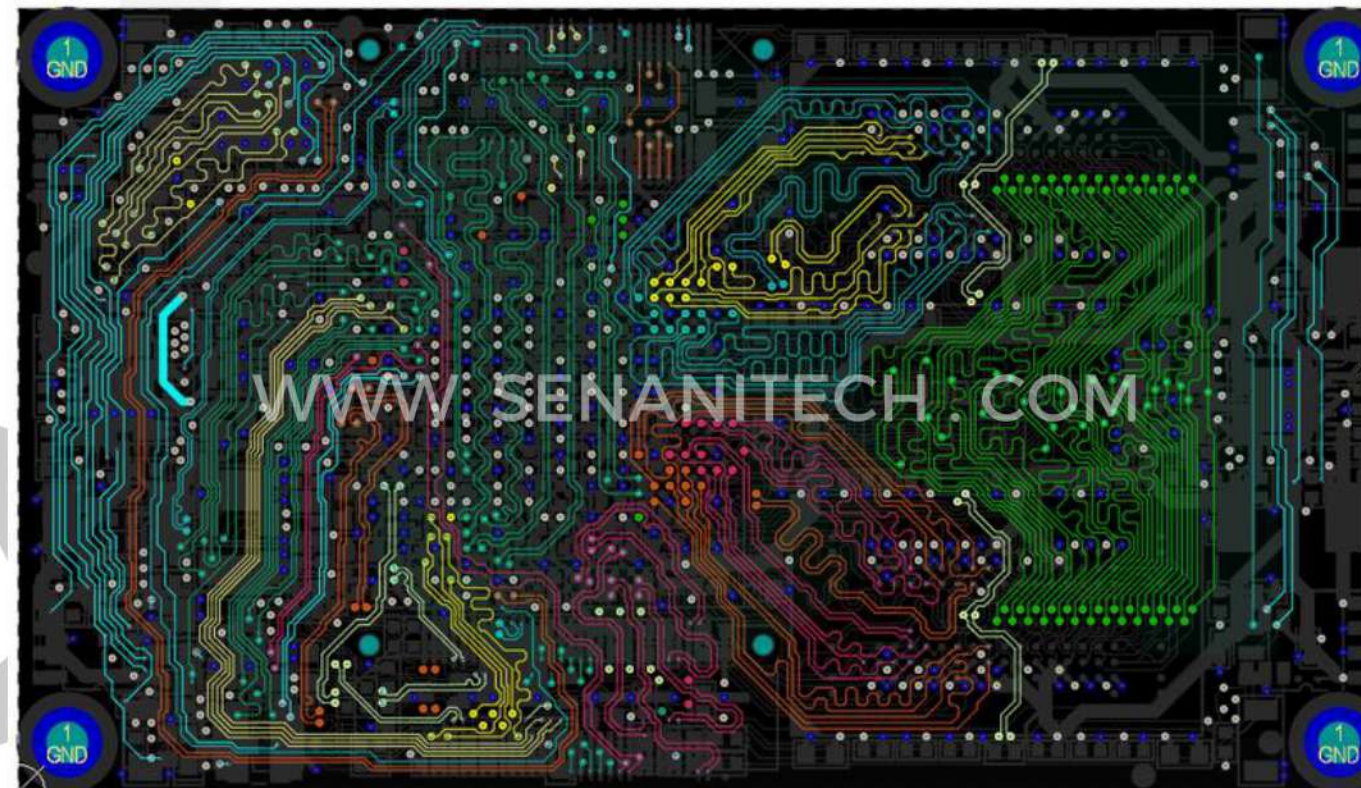
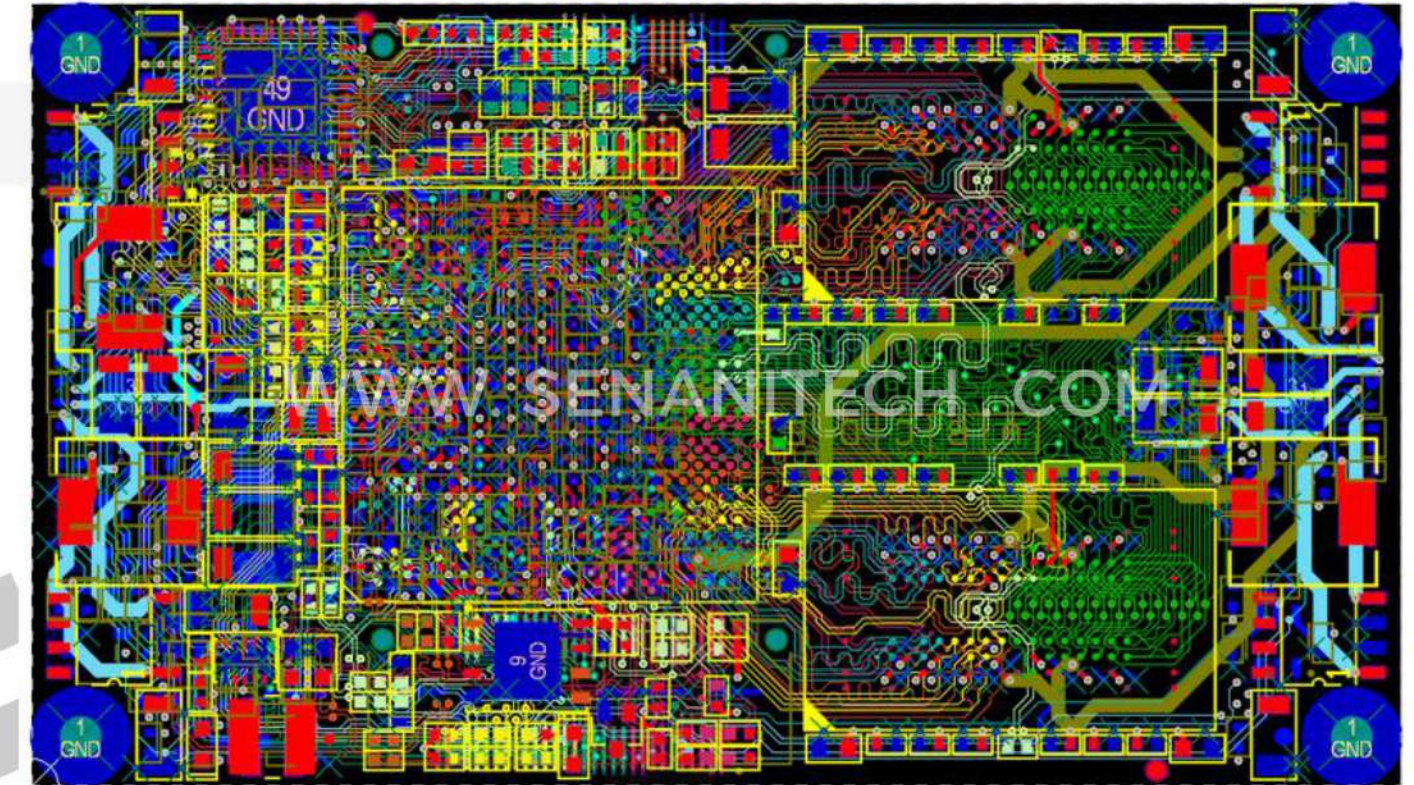
Aerospace PCBs

- HighDensity–RF Design
- HighLayercount–8-14 layers
- ImpedanceControlled



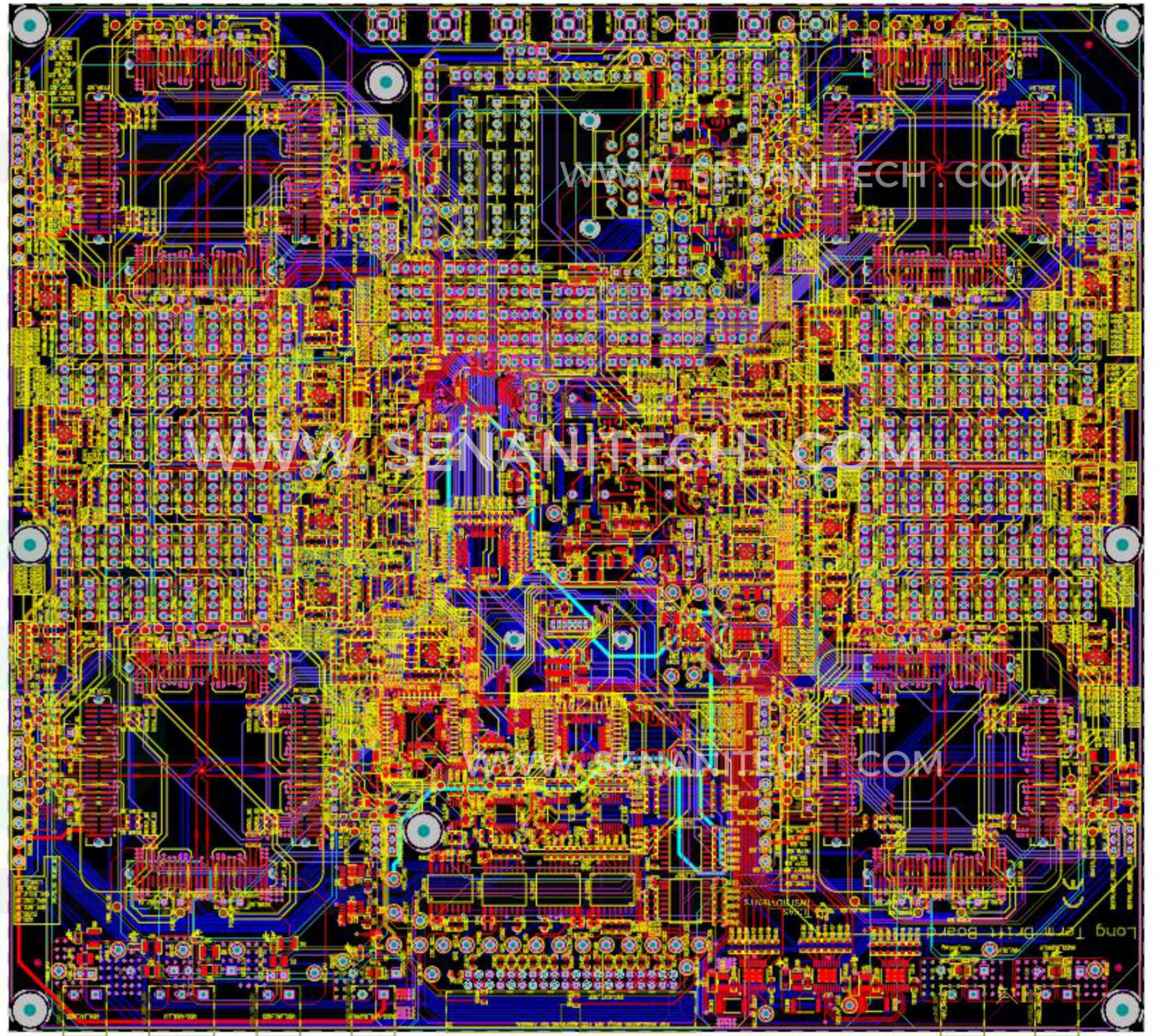
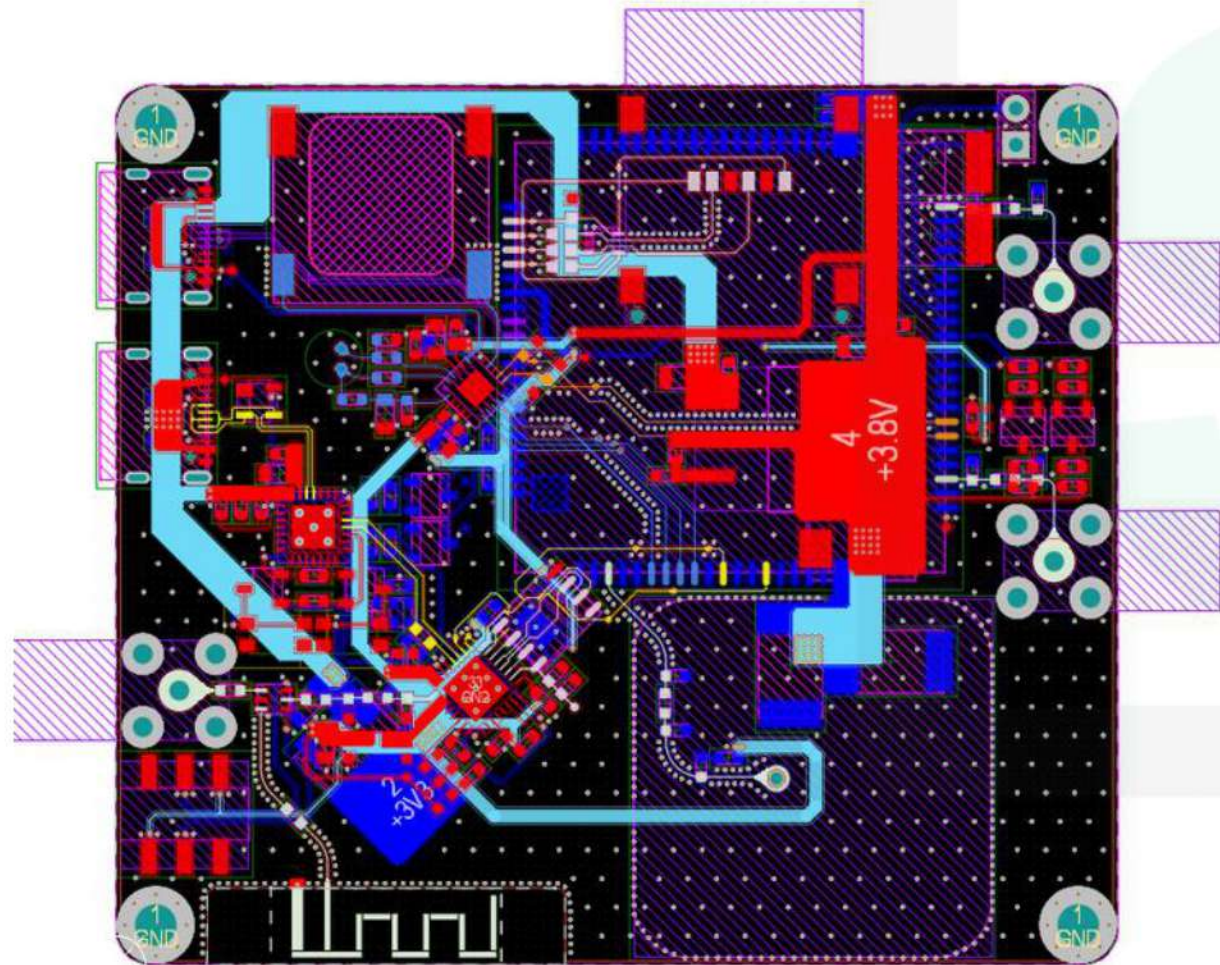
iMX6 Processor

- HighDensity-0.4mmBGA
- High Layercount-12 layers
- Impedance Controlled
- Dual DDR3

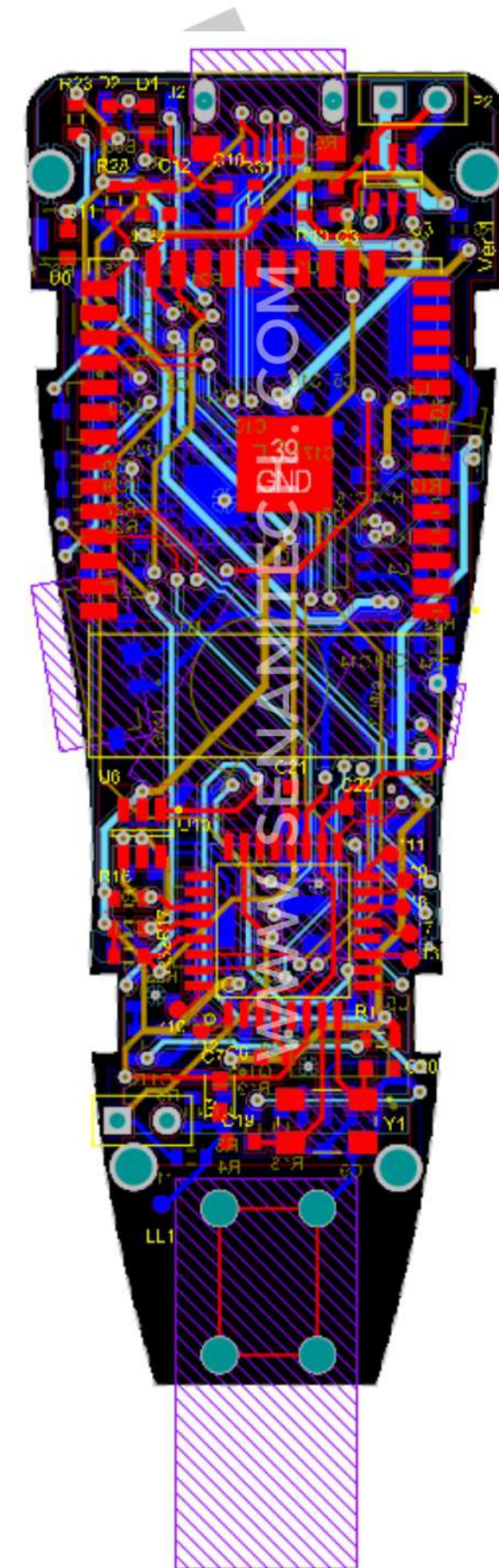
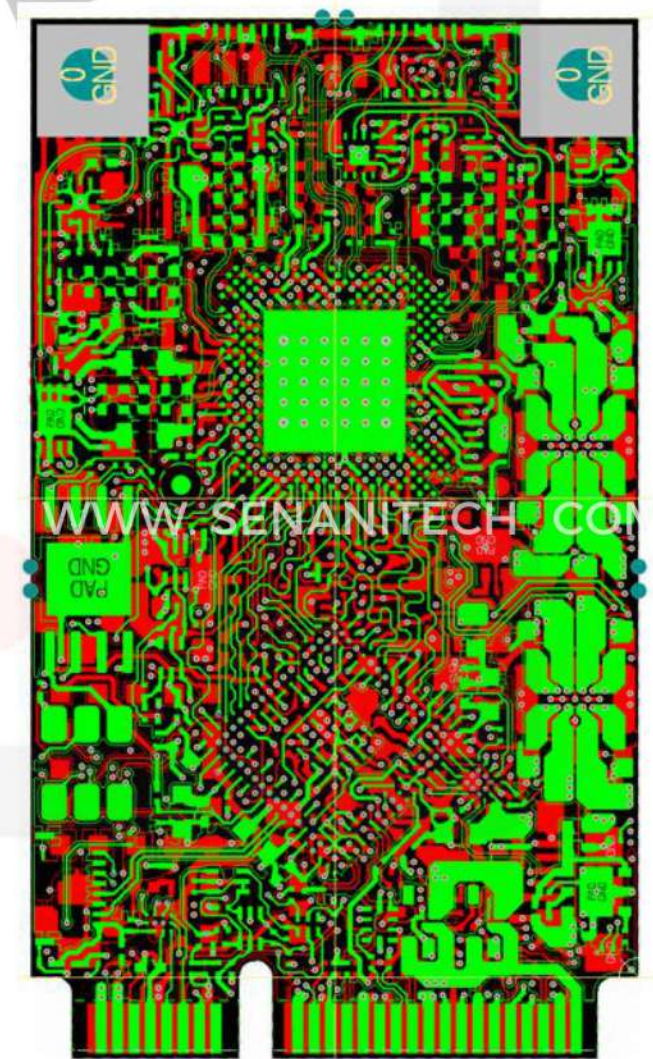
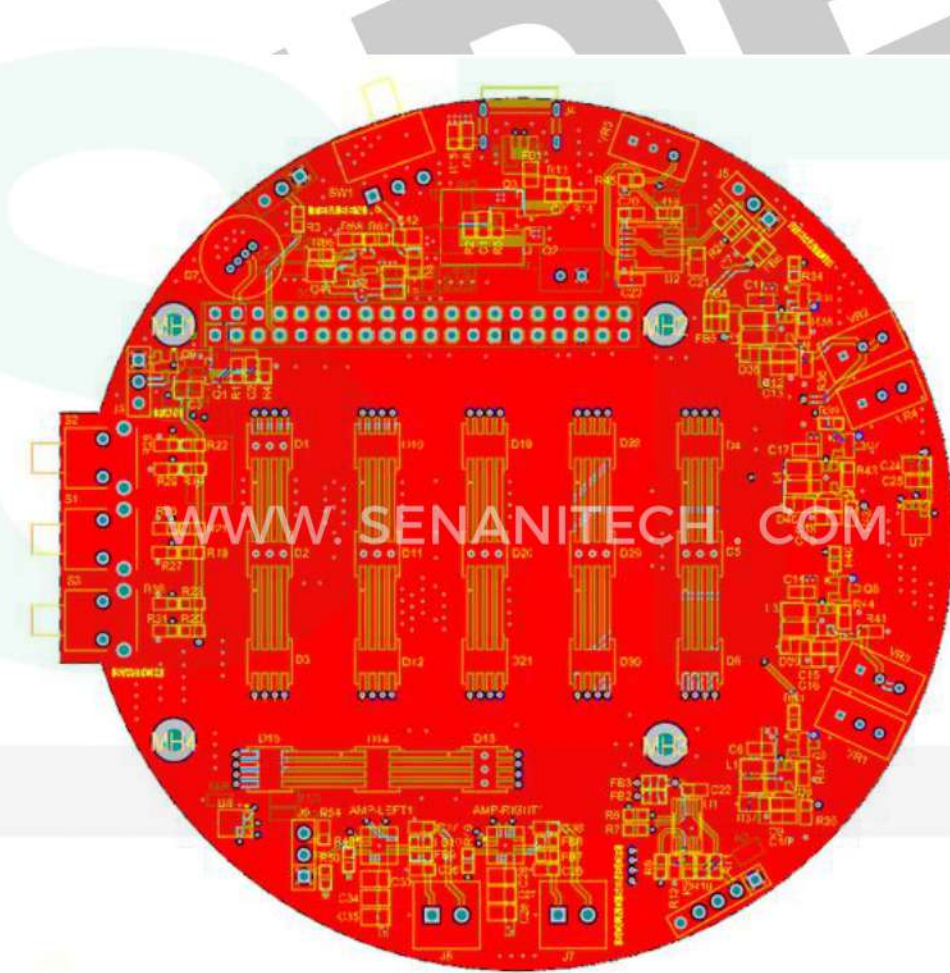
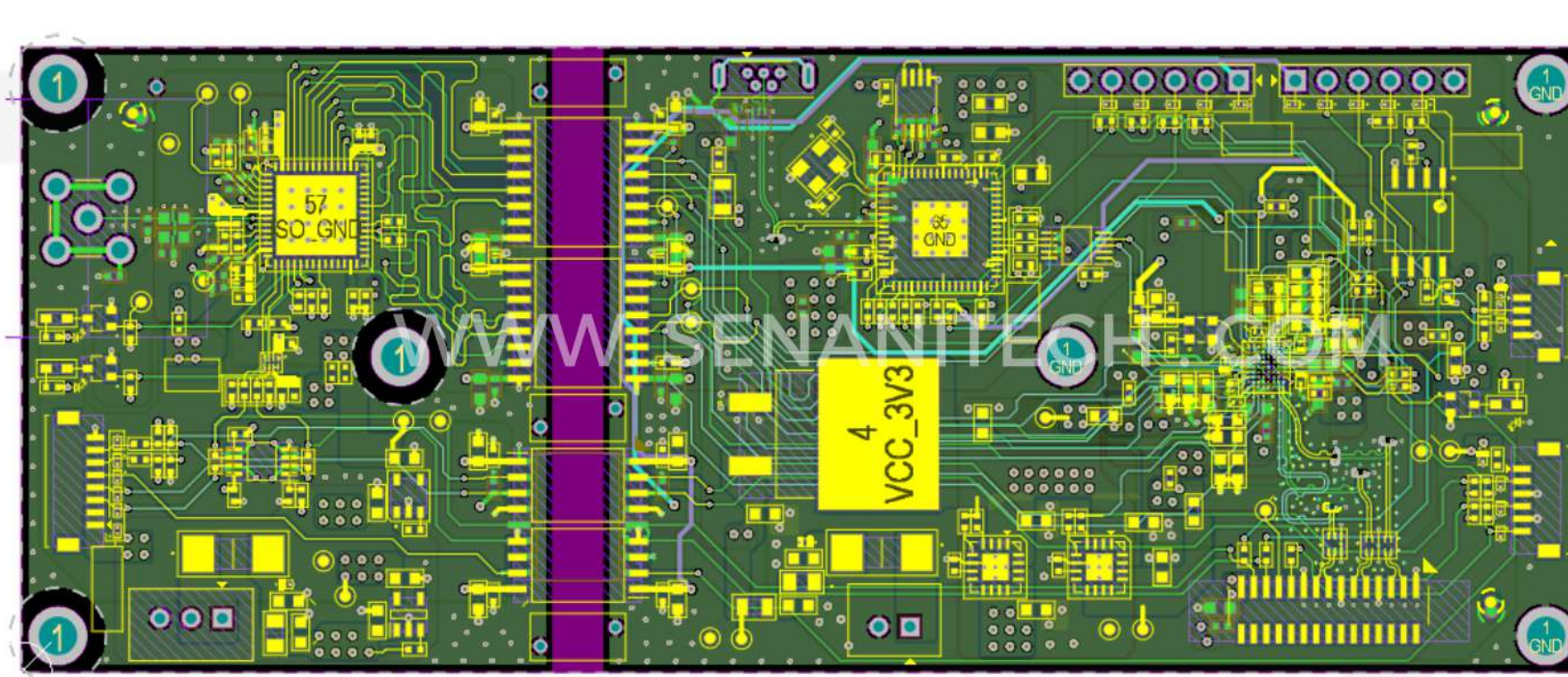
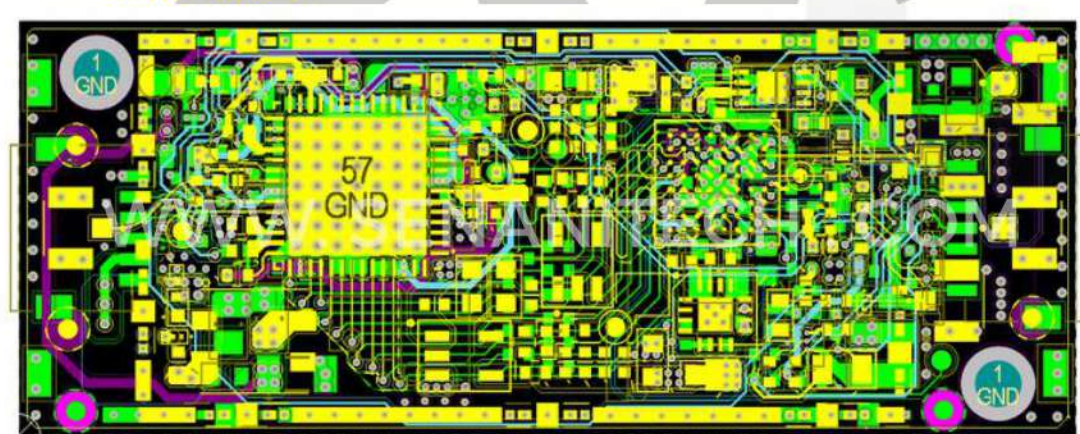
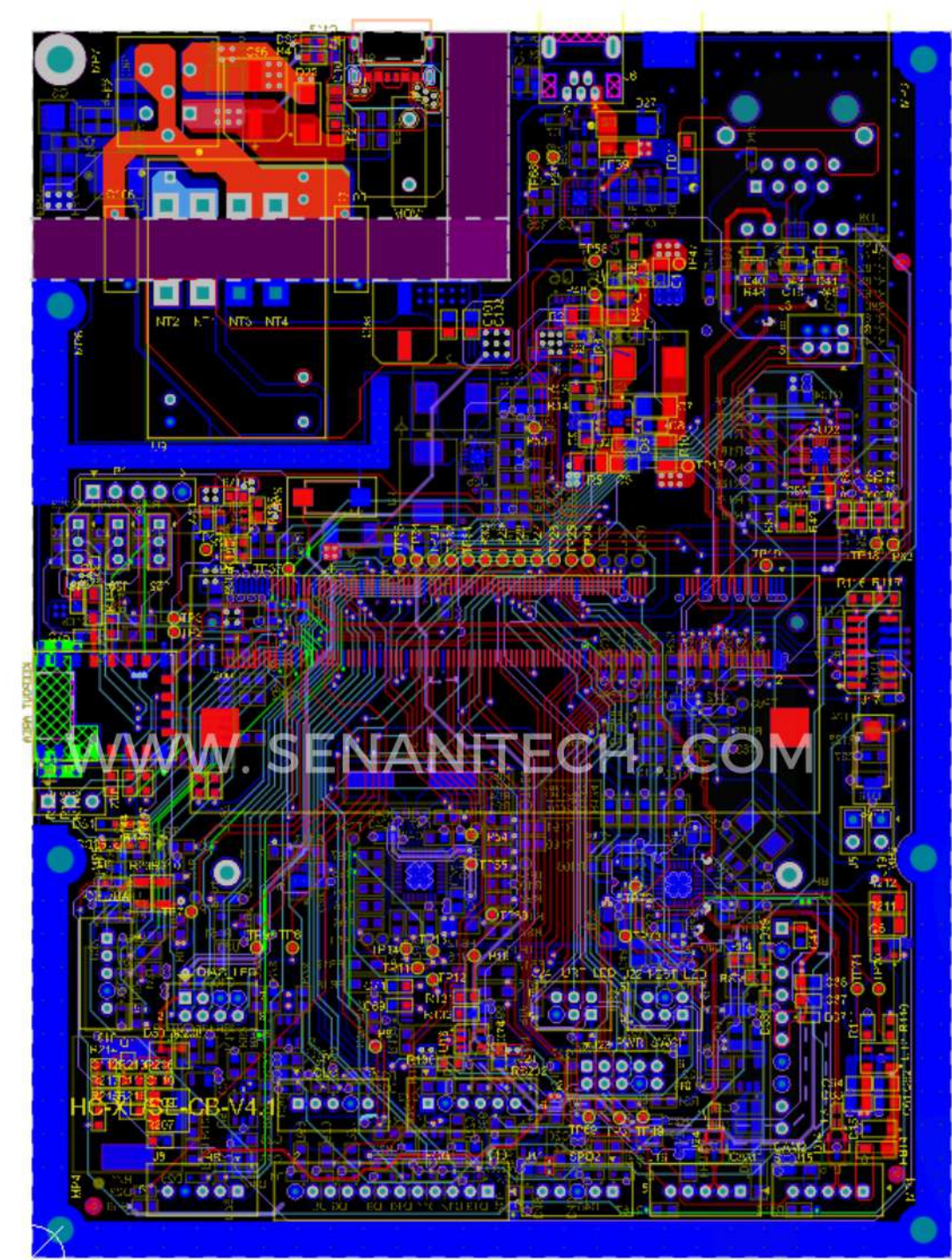


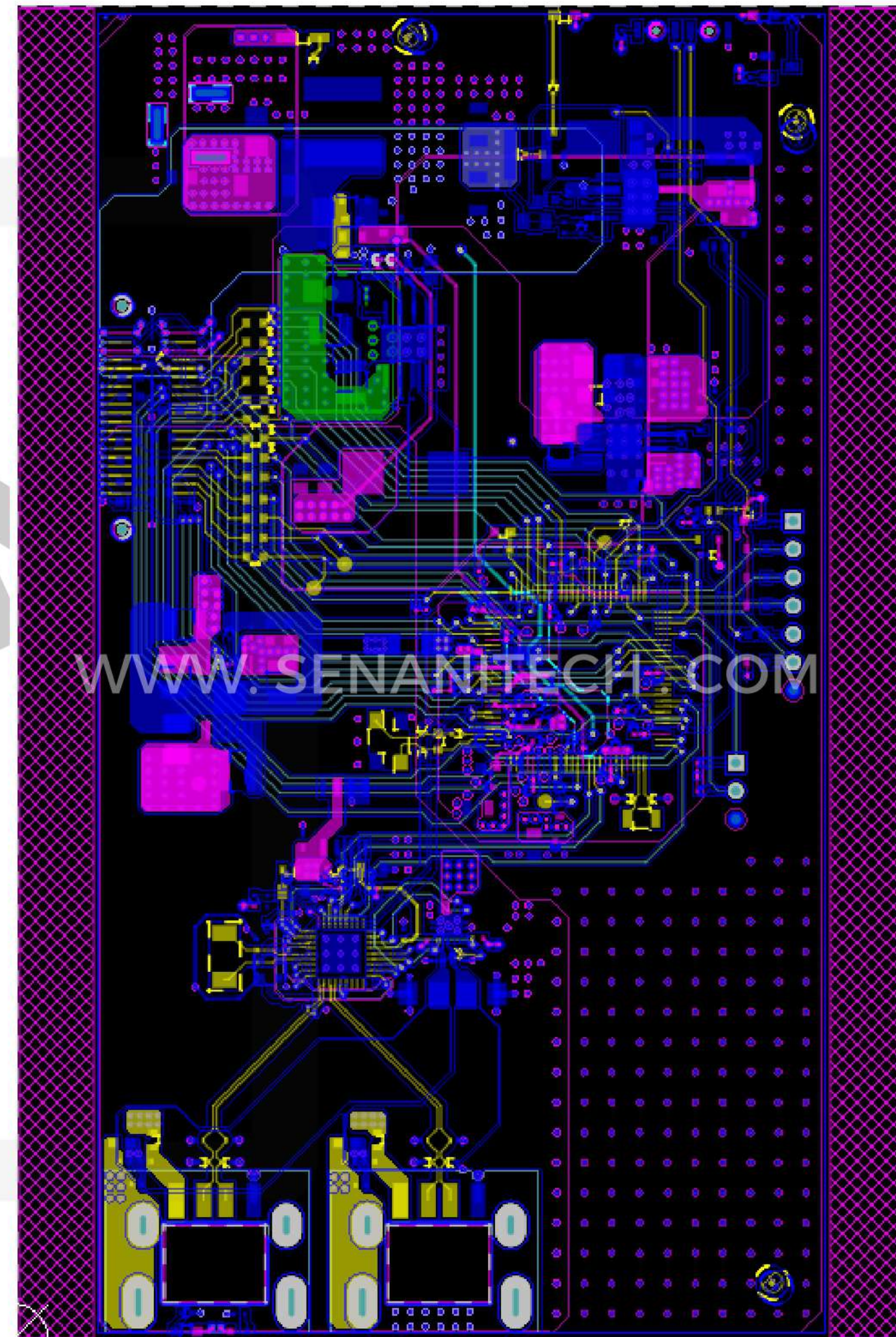
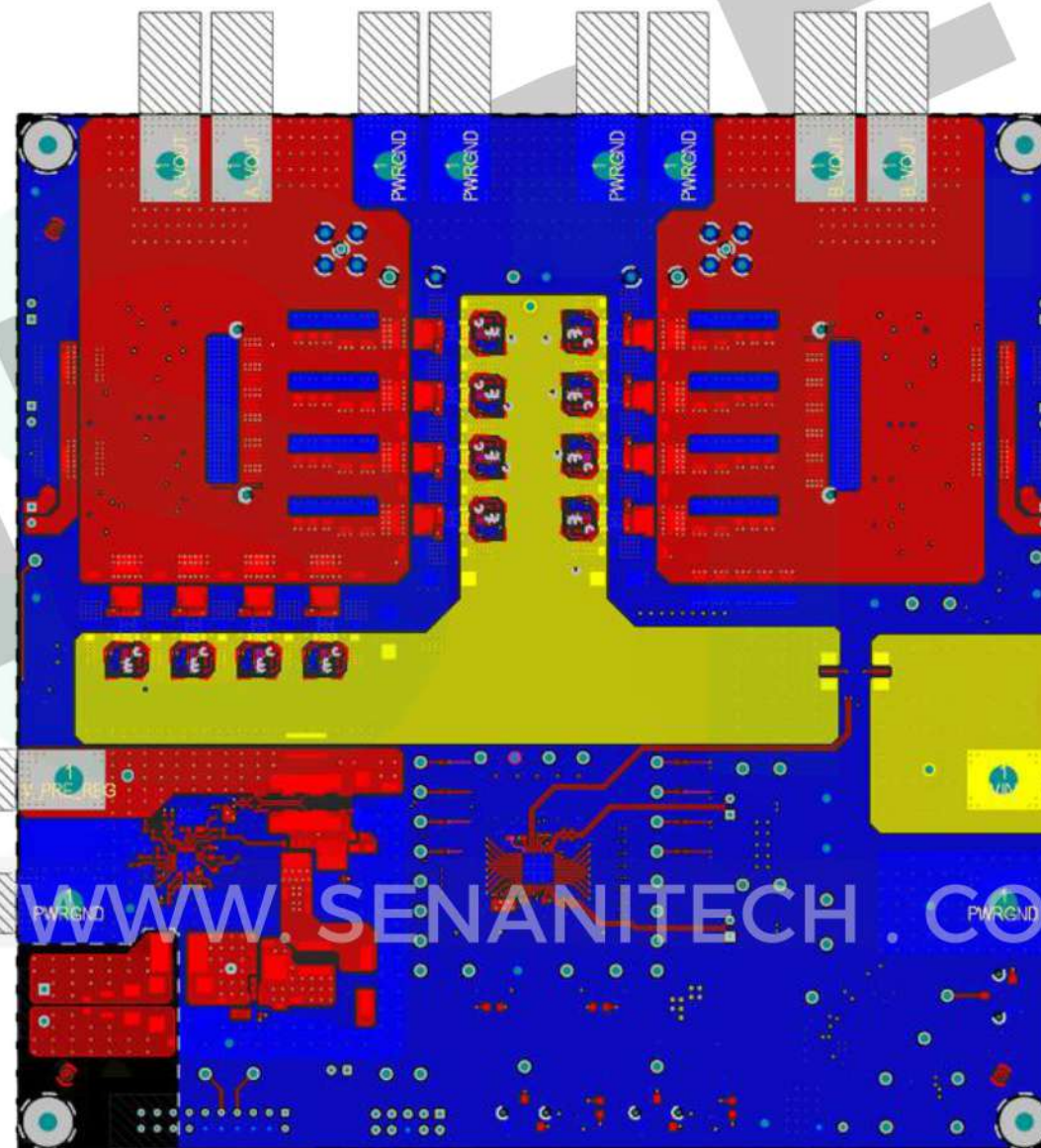
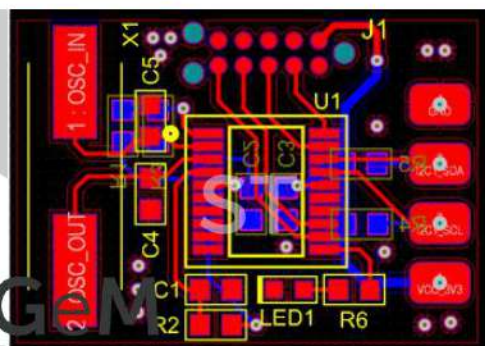
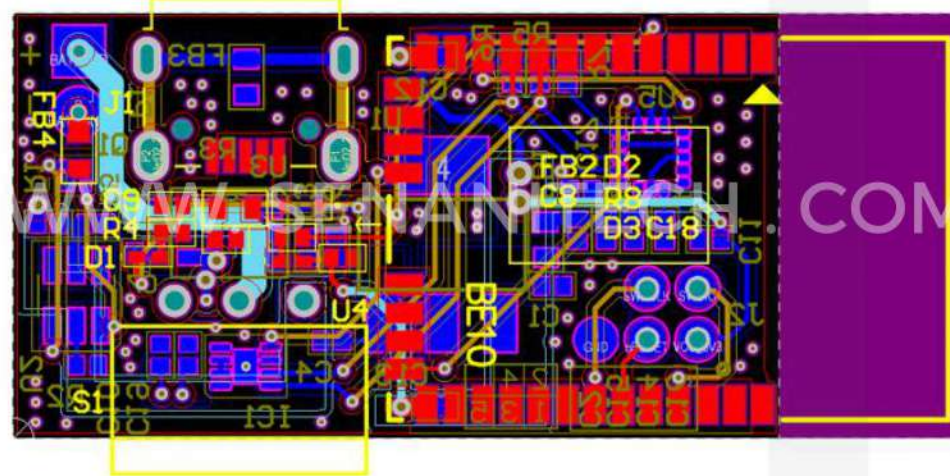
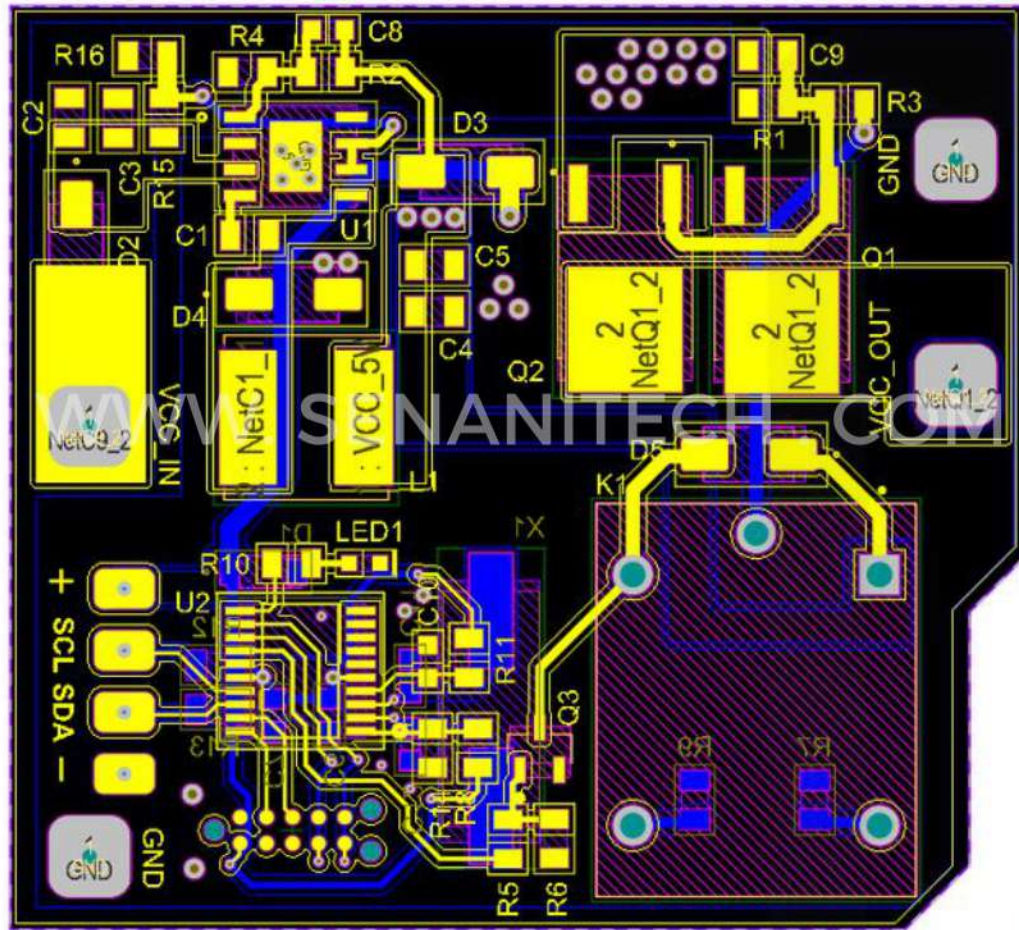
Industrial Control Card

- HighDensity Board
- HighLayercount-14 layers
- Impedance Controlled
- Controller Based



Medical Electronics PCBs





**ELECTRONICS
MANUFACTURING
SERVICES**

WWW.SENANITECH.COM

KEY FEATURES IN SOLDER PASTE PRINTER (MODEL : DEK 03I)

- Cycle time: 12 Sec (Excluding printing time)
- Applicable stencil size: L736mm x L736mm.
- Graphical Layer Base alignment function
- Stencil Vacuum Function: A stencil vacuum mechanism delivers consistent high accuracy printing with no effects from stencil drops.
- ASM Adjustable Stencil Mount
- PCB Size: L580x W508mm.
- Squeegee Pressure Mechanism: Software-controlled, motorised



Key Features in Pick and Place (Model : YSM10)

- Fixed plate: Max. 96 types (conversion for 8 mm tape feeder)
Tray: 15 types (maximum when equipped with sATS 15, JEDEC)
- Rated Speed: HM head (10 nozzle) specification : 46,000CPH
, HM 5 head (5 nozzle) specification : 31,000CPH
- PCB Size : L 510 x W 460mm
- Accuracy: +/-0.035mm (+/-0.025mm)
- Component range : 03015mm to Board W55 x L100mm
- Improves scan camera performance that expands the size of high-speed mounting parts to 12mm. Supports ball recognition such for BGA and CSP, etc.



Key features in Reflow oven (Model: KONARK 710S)



GeM
Government
e Marketplace



**MAXIMUM PCB WIDTH OF
400 MM**



**14 INDEPENDENTLY PID
CONTROLLED HEATING
ZONES**



MAX TEMP : 360°C



LEADFREE CAPABLE



**REPEATABILITY: - +/- 1°C
Mass Production: +/- 2°C**

WHY choose us



GeM
Government
e Marketplace



Skilled Engineers

Our highly skilled PCB designers deliver precise, reliable, and efficient designs, ensuring top quality and performance for every project.



Documentation

High-Quality PCB Design, Clean Documentation, Guaranteed Lower Prices Our team delivers exceptional PCB designs with thorough documentation, all at a guaranteed competitive price.



Organize

Well-organized schematic symbols, PCB footprints, and 3D models for efficient design integration. Ensuring accuracy and consistency throughout the entire process.



Proper Communication - Daily Status Report

We maintain a clean communication model, providing daily status reports and facilitating online design reviews for seamless collaboration.



24/6 Comprehensive Support

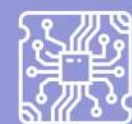
Our team provides round-the-clock support to ensure your design timeline is fully accommodated and compensated.



Discounted Rates for Long-Term Partnerships

We offer competitive discounts for clients engaging in long-term collaborations.

VALUE ADDED SERVICES



PCB Fabrication

2-80Layer / IPCClass2 and Class3 / HDI (2-n-2), (3-n-3)/ Flex/ Rigid - Flex
Component Sourcing



PCB Assembling / SMT/X-Ray/MicroBGA

We provide comprehensive PCB assembly services, ensuring high-quality production and efficient turnaround times for all your electronic projects.



Box Building and Power-On Testing Services

Our box building and power-on testing services ensure your assembled PCBs are securely packaged and fully functional, providing reliable performance right out of the box.



Strong Supplier Partnerships Across the Globe

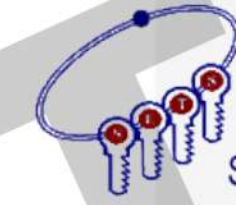
PatternPCB has established long-term relationships with suppliers in the USA, Taiwan, China, and India. We can help you select the right supplier to meet your specific needs.

VALUABLE CLIENTS

BYNET
Testing Systems



NEUROSTELLAR



SETS

Society for Electronic Transactions and Security



HealthCubeTM

SMART DIAGNOSTICS STARTS HERE



CurneuTM

Care beyond reach



VOLTECH

EXCELLENCE IN ENGINEERING PRODUCTS & SERVICES

kirloskar



Thank
you!

ENGINEERING DELIVERY CENTRE

2ND FLOOR, SRI VALLIAPPA
BLOCK, SONA INCUBATION
FOUNDATION, JUNCTION MAIN ROAD
SALEM - 636005, TAMIL NADU, INDIA

SENANITECH - MANUFACTURING

INFITECH ELECTRONICS NO:15
CORPORATION ROAD SEEVARAM
PERUNGUDI CHENNAI -600096

WWW.SENANITECH.CO

M